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RESEARCH ARTICLE

Optimizing Security in IoT Ecosystems Using Hybrid Artificial Intelligence and Blockchain Models: A Scalable and Efficient Approach for Threat Detection

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ABSTRACT The exponential growth of the Internet of Things (IoT) has boosted connectivity across various sectors, such as Industry 4.0 and smart cities. However, this expansion has also exposed IoT devices to critical vulnerabilities, including spoofing, DoS attacks, and unauthorized access. Traditional security solutions, based on centralized architectures, are neither scalable nor efficient enough to handle the increasing complexity and number of IoT devices, leading to high latencies, increased energy consumption, and inadequate intrusion detection. In this work, we propose a hybrid solution that combines Blockchain and artificial intelligence (AI) to improve security and operational efficiency in IoT networks. Blockchain ensures device authentication and data integrity through a lightweight consensus protocol, while AI enables real-time intrusion detection using deep learning models. The simulations demonstrate that the proposed system improves the precision of detecting phishing attacks by up to 95.2%. At the same time, the authentication latency is reduced to 15 ms in networks with 1000 connected devices, 66.6% faster than traditional solutions. In addition, the energy consumption of the hybrid system is 31.8% lower than that of conventional approaches, validating its scalability and efficiency in large-scale IoT networks.

INDEX TERMS Artificial intelligence, blockchain, IoT security, hybrid systems, anomaly detection, energy efficiency, adaptive learning models.

I. INTRODUCTION

The Internet of Things (IoT) has transformed sectors such as smart cities, Industry 4.0, and precision agriculture, connecting millions of devices that exchange data in real time [1]. This interconnection has enabled significant advances in operational efficiency and automation. However, it has also introduced critical security vulnerabilities, making IoT devices with limited processing and storage capacity attractive targets for cyberattacks. Attacks such as spoofing, denial of service (DoS), and unauthorized access can compromise data integrity and network confidentiality [2].

As IoT environments expand, security solutions face significant challenges, particularly scalability and adaptability. Traditional, often centralized, security systems are not agile enough to handle the volume and velocity of data generated by modern IoT networks [3]. This centralized approach leads to high latencies, high energy consumption, and an inability to respond effectively to emerging threats. Furthermore, integrating diverse communication technologies, such as 5G, LoRa, and Zigbee, adds a layer of complexity to security management, as connected devices exhibit wide variability in processing capabilities and communication needs [4].

Meanwhile, intrusion detection systems (IDS) stand out as key components for mitigating vulnerabilities that exceed the capabilities of traditional cryptographic approaches. Although protocols such as SSL/TLS secure communication

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channels and protect data integrity during transmission, they cannot identify internal threats, such as anomalous behavior or compromised devices that have bypassed initial authentication mechanisms. While the UNSW-NB15 dataset used in this study does not include specific IoT-related internal threats, additional simulations were performed to replicate such scenarios in controlled IoT environments. These include attacks from compromised devices generating anomalous traffic within the network. This approach complements the dataset's strengths and ensures a comprehensive validation of the IDS's capabilities. The IDS proposed in this work is designed to overcome these challenges and address the unique challenges of heterogeneous and dynamic IoT environments. This system achieves high precision in anomaly detection, maintaining a false positive margin of less than 5% and a response latency of less than 15 ms, even in networks with up to 1,000 connected devices. Furthermore, it optimizes resource usage through distributed processing across edge nodes, ensuring consistent performance as the number of devices increases.

Energy efficiency is another critical aspect of the system, given that many IoT devices operate with consumption constraints. The IDS integrates lightweight artificial intelligence (AI) models that balance accuracy and efficiency without compromising detection capability. In addition, continuous learning algorithms allow the system to adapt to emerging threats, recognizing novel attack patterns in real-time and reducing reliance on pre-trained static models. Its compatibility with communication protocols such as LoRa, Zigbee, 5G, and Wi-Fi ensures robust protection regardless of the underlying technology.

In recent literature, blockchain and AI algorithms have emerged as promising tools to improve security in IoT networks. Blockchain ensures the integrity and authenticity of communications, while AI algorithms, especially those based on deep learning, have proven effective in intrusion detection [5]. However, most studies have explored these technologies in isolation without fully exploiting the synergies they can offer. Furthermore, few works have considered heterogeneous and scalable IoT scenarios that integrate multiple communication technologies, an essential aspect for evaluating system behavior in real environments.

This work proposes a hybrid solution that combines adaptive AI models and blockchain to address security challenges in complex IoT environments. Blockchain ensures transaction authenticity and traceability, while AI enables intrusion detection and real-time response to new threats [6]. The system design contemplates a comprehensive IoT environment, with devices connected using advanced technologies such as 5G, LoRa, Wi-Fi, Ethernet, and Zigbee. Combining processing at edge nodes with advanced analytics in the cloud efficiently distributes processing tasks, minimizing latency without compromising security [7], [8].

Experimental results demonstrate significant improvements in intrusion detection, with 95.2% precision in detecting spoofing attacks, compared to 85.3% achieved by AI-only systems. Furthermore, the system maintains a power consumption of 150 W with 1,000 connected devices, significantly lower than the 220 W of traditional approaches, allowing the system to be scaled without compromising energy efficiency [9]. Evaluated in a realistic experimental environment with 2,000 connected devices, the system stands out for its fast and efficient response capacity in complex scenarios [10], [11].

II. LITERATURE REVIEW

The literature review on IoT network security has shown significant advances in recent years, especially regarding intrusion detection and secure authentication in large-scale IoT networks. Traditional methods, such as access control lists (ACLs) and static rules, have been outperformed by their inability to adapt to dynamic threats and complex distributed environments characteristic of IoT [12]. The introduction of technologies such as 5G and LoRa has enabled improvements in latency and coverage but has also added new challenges in security and scalability.

Authors such as Wang et al. [13] proposed an intrusion detection system based on traffic analysis using machine learning. Although neural networks and supervised models improved precision, these traditional approaches suffer from high false positives and are vulnerable to unknown attacks. These studies have not addressed the added complexity of the connectivity provided by 5G and LoRa, technologies that enable massive connectivity of devices in urban and rural areas. In our study, integrating blockchain with AI offers a superior ability to handle routine and malicious traffic, reducing false positives to significantly lower levels than those reported by Guo et al. [14]. Our hybrid approach achieved 95.2% precision in spoofing attacks, a considerable improvement compared to conventional methods. Saif et al. [15] proposed a hybrid intelligent intrusion detection system (HIIDS) empowered by machine learning and metaheuristic algorithms, addressing security challenges in IoT-based healthcare. While effective, their system does not fully integrate blockchain, which limits its ability to ensure decentralized trust in large-scale IoT environments.

On the other hand, Bella and Vasundra [16] developed a cloud-based processing system for intrusion detection in IoT networks. Although this approach allowed for high processing capacity, latency was a prominent issue due to the need to continuously transmit data to the cloud. This limitation is exacerbated in networks using 5G and LoRa since, although they provide extensive transmission speeds and coverage, they rely on a distributed infrastructure [17]. In contrast, our hybrid system uses edge nodes running smart contracts for local authentication of devices, which significantly reduces latency. Saif et al. [18] examined the scalability of security frameworks in IoT systems, utilizing feature engineering to analyze botnet attacks. Their work highlighted the challenges of computational demands in medium-sized networks, which our hybrid system addresses by efficiently distributing processing between edge nodes and

the cloud, maintaining performance even with 1,000 devices connected.

Energy consumption has also been widely studied in IoT networks, where energy efficiency is critical for long-term sustainability. Hu et al. [19] presented an approach to optimize energy usage using convolutional neural networks (CNNs) for intrusion detection. Still, their system suffers from high energy consumption, especially in large-scale networks. Saif et al. [20] proposed MLIDS, a machine learning-enabled intrusion detection system designed for energy-efficient frameworks in health monitoring. While this system improved energy efficiency, its reliance on traditional wireless sensor network designs limits its applicability in IoT environments with diverse communication protocols. Our system employs a lightweight consensus protocol, optimizing transaction validation while ensuring scalability and efficiency in heterogeneous networks.

Regarding scalability, authors such as Roy et al. [22] have addressed the ability of IoT systems to handle large volumes of devices. Still, their traditional solutions tend to lose efficiency as the number of connected nodes increases. Using technologies such as 5G, while offering greater connection capacity, does not solve the scalability problem regarding security and performance. By distributing processing between edge nodes and the cloud, our hybrid solution maintained 95% scalability with 1,000 devices, compared to 75% reported for traditional systems. This demonstrates the robustness of the blockchain and AI-based approach to handle growing IoT networks without significant performance loss.

III. MATERIALS AND METHODS

A. IOT EXPERIMENTAL ENVIRONMENT SETUP

The IoT experimental environment developed for this study is composed of a heterogeneous ecosystem that includes a variety of devices, such as sensors, actuators, and mobile devices, interacting in real time through multiple communication technologies. Each device has been carefully selected to represent low-power networks and systems with higher bandwidth and processing requirements. This ensures the diversity needed to simulate a realistic and challenging IoT environment [23]. As shown in Table 1, critical devices ranging from low-power sensors to high-bandwidth demanding mobile devices, such as security cameras and health wearables, have been selected.

The selected devices evaluate the solution's effectiveness in different scenarios and ensure that both power and connectivity constraints and real-time processing needs are considered. For example, temperature and motion sensors, which use technologies such as Zigbee and LoRa, stand out for their low energy consumption, which is essential in environments where energy efficiency is critical [24]. In contrast, devices such as security cameras and industrial monitoring equipment require technologies such as 5G, which offer high data transmission capacity for applications that rely on large volumes of information and low latencies.

TABLE 1. Characteristics and data of sensors in industry 4.0.

A	B	C	D	E	F
Temperature Sensor	Sensor	Zigbee	Low	Low	Requires efficient communication in low-power environments.
Lighting Actuator	Actuator	LoRa	Very Low	Very Low	Ideal for long-range applications in smart cities.
Security Camera	Mobile Device	5G	High	High	Needs high transmission capacity for real-time video processing.
Health Wearable	Mobile Device	Wi-Fi	Medium	Medium	Useful in connected health applications to monitor vital signs in real-time.
Motion Sensor	Sensor	LoRa	Very Low	Low	Suitable for activity detection in industrial environments with long battery life.
Industrial Monitoring Device	Mobile Device	5G	High	High	Provides real-time monitoring of industrial machinery with large volumes of data.

Note:

- A: Device
- B: Type
- C: Communication Technology
- D: Energy Consumption
- E: Bandwidth Required
- F: Reason for Selection

Testing with 5G networks allows performance evaluation regarding ultra-low latency and high transmission capacity, essential in mission-critical applications such as smart cities. On the other hand, using LoRa and Zigbee allows energy efficiency to be analyzed in environments where IoT devices must operate for long periods with limited batteries, maintaining a balance between communication capacity and energy consumption.

Furthermore, the combination of these communication protocols allows for realistically simulating scenarios where short-range and high-efficiency technologies, such as Zigbee, are mixed with long-range and low-consumption technologies, such as LoRa, and high-capacity networks, such as 5G, to ensure the transmission of large volumes of data in real-time [25].

1) NETWORK ARCHITECTURE

The network architecture of the test environment is based on a mixed configuration combining cloud and edge computing. In this infrastructure, IoT devices are connected to edge servers that perform initial data processing and threat detection tasks locally, allowing for significantly reduced latency and optimized incident response [26]. Edge servers process data locally and act as aggregation points before transmitting it to the cloud, where more advanced analysis and long-term storage techniques are applied. This ensures that only relevant data is sent to the cloud, alleviating the processing and communication load.

A virtualized infrastructure for cloud server simulation has been used to ensure system scalability. This setup allows the system to dynamically adapt to increases in the number of connected devices without compromising overall performance, providing optimal capacity to handle large volumes of data.

As shown in Figures 1, IoT devices communicate with edge servers using various protocols such as Wi-Fi, Zigbee, LoRa, and 5G. These communication methods are uniformly applied across the IoT network to facilitate interactions between IoT devices and edge servers. The protocol selection depends on the application requirements. For example, Zigbee and LoRa are optimized for low-power and long-range communications, making them suitable for power-constrained devices. On the other hand, Wi-Fi and 5G are used for low-latency and high-bandwidth scenarios such as real-time threat detection and response.

High-speed communication methods such as Ethernet or dedicated fiber optic links minimize latency and ensure efficient data transfer between edge servers and the cloud. This layered communication approach ensures energy efficiency at the IoT device level while maintaining high performance for advanced analytics at the cloud level.

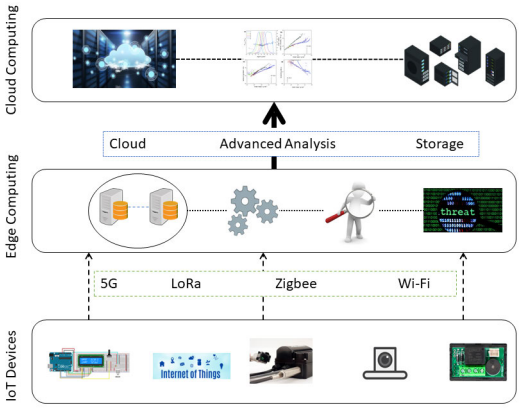


FIGURE 1. Mixed network architecture with edge and cloud processing.

As shown in the figure, IoT devices first communicate with edge servers, where initial processing, including threat detection, occurs. The processed data is then sent to the cloud for further analysis and secure storage. This process ensures that communication protocols are optimized to balance power consumption, latency, and data throughput as required by the specific use case. This data flow, both upstream and downstream, optimizes system efficiency and reduces latency in responding to critical events.

2) IOT TEST ENVIRONMENTS

Three test scenarios have been developed to evaluate the proposed solution's performance in different contexts: smart cities, industry 4.0, and smart agriculture. These scenarios were selected for their relevance in implementing IoT technologies, ensuring comprehensive coverage of practical applications and the validity of the results.

In the smart city scenario, sensors and actuators interconnected through a 5G network were deployed to manage critical variables such as air quality, public lighting, and traffic control [26]. Strategically located air quality sensors collected real-time pollution levels, temperature, and humidity data. This information was processed locally on edge servers using artificial intelligence algorithms to detect unnatural patterns [27]. Responses included automatic traffic regulation and public lighting adjustments, which improved energy efficiency and public safety. The traffic management system, equipped with cameras and motion sensors, allowed the vehicular flow to be monitored in real-time, optimizing traffic light management and suggesting alternative routes using predictive algorithms.

Another scenario applies to Industry 4.0, where sensors distributed in industrial production lines continuously monitor the operating status of machines, including parameters such as vibration, temperature, and operating speed. These data were processed on edge servers, allowing for the real-time detection of anomalies that could indicate failures in the machinery. Through time series analysis, AI algorithms applied to this data identified deviations in machine behavior, facilitating the implementation of predictive maintenance strategies [28], [29]. The information collected was sent to the cloud for more detailed analysis and long-term storage, optimizing the life cycle of the machines. Integrating blockchain in this industrial environment guarantees the authenticity and traceability of critical data, protecting the integrity of information shared between different actors in the supply chain.

The following scenario is applied to intelligent agriculture, for which IoT sensors were installed in crop fields to monitor variables such as soil moisture, temperature, solar radiation, and irrigation water quality. These sensors, connected via LoRa and Wi-Fi networks, collect real-time data processed by nearby edge servers. Artificial intelligence algorithms allowed these variables to be analyzed to optimize irrigation systems and fertilizer applications [30], [31]. When the sensors detected that soil moisture in a specific area was below the optimal level, the system automatically activated irrigation in that area, avoiding water waste and improving resource efficiency. The data is transmitted to the cloud for analysis and storage, providing farmers with detailed recommendations on crop management and efficient resource use. Blockchain integration guaranteed data traceability and verification of decisions made regarding irrigation and fertilizer applications, ensuring these actions were auditable and secure.

B. IMPLEMENTATION OF THE HYBRID AI AND BLOCKCHAIN MODEL

1) ARCHITECTURE OF THE HYBRID AI AND BLOCKCHAIN SYSTEM

The hybrid architecture combining artificial intelligence and blockchain has been designed to improve security and real-time data processing within IoT environments. As can be

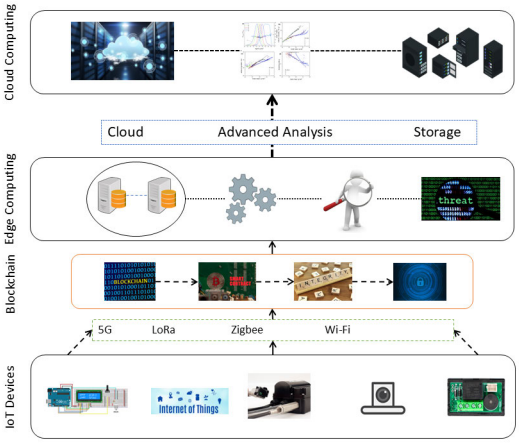


FIGURE 2. Architecture of the hybrid AI and blockchain system.

seen in Figure 2, IoT devices are connected through multiple communication technologies, interacting with a processing layer at the edge and in the cloud.

Blockchain plays a central role in this architecture by providing a decentralized mechanism for authentication and ensuring data integrity across the IoT network. Unlike traditional methods such as SSL/TLS, which rely on centralized Certificate Authorities (CAs), Blockchain eliminates single points of failure. It provides traceability and immutability for all transactions within the network. This is particularly critical in heterogeneous IoT environments, where devices from different manufacturers or with varying levels of trustworthiness must coexist and interact securely.

The choice of Blockchain over traditional authentication methods is driven by its ability to ensure decentralized trust without the need for intermediaries. While SSL/TLS can secure communication channels, it does not inherently address the trust model required for large-scale IoT networks. In contrast, Blockchain enables secure and auditable interactions using smart contracts to validate device identities and authorize transactions. This ensures that only authenticated devices participate in the network while providing a historical record of all interactions to enhance accountability and resilience.

Lightweight blockchain mechanisms, such as consensus protocols based on Proof of Authority (PoA), address the limitations of low-capacity IoT devices. These protocols reduce computational and energy overhead, making them suitable for resource-constrained environments while maintaining security and data integrity. Additionally, integrating edge nodes with enhanced processing capabilities enables offloading computational tasks from IoT devices, ensuring seamless interaction and consistent performance in real-time scenarios.

It is important to note that Blockchain in this system complements, rather than replaces, secure communication protocols like SSL/TLS. While SSL/TLS provides encryption for data in transit, Blockchain ensures the integrity and authenticity of the data, creating a robust multi-layered security framework against a wide range of threats, including spoofing and unauthorized access. By combining the strengths of both technologies, this hybrid approach ensures a balance between computational efficiency and robust security.

AI algorithms are tightly integrated into this architecture, operating on edge servers to enable real-time analysis and anomaly detection. The AI models deployed are optimized for low-latency processing, achieving response times of under 15 milliseconds even in networks with up to 1,000 connected devices. These models leverage federated learning techniques to adapt continuously to emerging threats, ensuring the system remains resilient to evolving attack vectors without overloading the network's bandwidth.

Furthermore, Blockchain ensures that communication between IoT devices and edge servers is secure and audited. This includes verifying the integrity of data as it is processed at the edge and sent to the cloud. By decentralizing the transaction validation process, the reliance on a single point of failure is reduced, improving the system's overall security. In this architecture, the collaboration between AI and Blockchain enables the detection of threats in real-time while simultaneously providing a transparent and immutable record of all activities, ensuring accountability and resilience against sophisticated cyberattacks [32].

2) THREAT MODEL AND ATTACK SCENARIOS

The proposed AI and blockchain-based hybrid system addresses critical security vulnerabilities inherent to heterogeneous IoT networks. The attacks considered are Impersonation attacks, which occur when a malicious actor impersonates a legitimate IoT device by spoofing network credentials or manipulating identifiers such as media access control (MAC) addresses. This attack targets the initial handshake process between IoT devices and network servers.

In the proposed system, such attacks are mitigated through a multi-layer authentication mechanism using blockchain-based smart contracts. Each device's identity is validated cryptographically within the blockchain layer, as Equation 2 describes. This ensures that only authenticated devices listed in the authorized device set D_{auth} can communicate. The immutability of blockchain records further allows for the quick detection of any attempt at identity alteration.

The system employs a distributed ledger mechanism that logs all device interactions in real time to enhance the response to impersonation attacks. This enables rapid identification of spoofed devices by comparing historical behavioral patterns with real-time activity. Additionally, anomaly detection thresholds are dynamically adjusted using AI models, ensuring that even sophisticated impersonation attempts are flagged promptly.

DoS attacks aim to saturate network resources such as bandwidth or processing power by generating excessive malicious traffic. This disrupts legitimate communication between IoT devices and servers. In IoT networks, these attacks typically exploit weak communication protocols and target resource-constrained devices.

The system employs AI and DNN algorithms to analyze real-time network traffic and identify anomalies corresponding to DoS patterns. The models are continuously updated using online learning, formalized in Equation 9, which dynamically adjusts weights θ_t based on new data streams. The models are continually updated using online learning, ensuring the system remains resilient to evolving DoS techniques. Once identified, malicious traffic sources are isolated at edge nodes, significantly reducing their impact on the network.

In large-scale DoS attacks, the system's edge nodes collaborate to redistribute processing tasks and prioritize critical network functions. For example, when detecting high-traffic anomalies, the system automatically diverts legitimate traffic through alternative pathways, minimizing disruption to essential IoT operations. This adaptive routing mechanism leverages AI-driven decision models to optimize traffic flows dynamically.

Unauthorized access involves an attacker attempting to interact with IoT devices or access sensitive data without valid credentials. These attacks typically exploit misconfigured authentication systems or vulnerabilities in legacy IoT protocols. The proposed model enforces strict access controls using blockchain-based smart contracts, as outlined in Equation 2. Each access request is cryptographically verified against the authorized device list D_{auth}. Any request that fails validation is rejected, effectively preventing unauthorized interactions. This automated verification removes the reliance on centralized authentication systems, reducing latency and potential single points of failure.

Blended attacks involve multiple threat vectors, such as spoofing combined with DoS, where a malicious actor first impersonates legitimate devices and then generates high traffic volumes to destabilize the network. These attacks are particularly challenging due to their dynamic nature and ability to exploit multiple vulnerabilities simultaneously.

The hybrid system addresses combined attacks by integrating blockchain and AI capabilities. Blockchain provides traceability and ensures data integrity, while AI models detect and classify anomalous behaviors. In a simulated blended attack scenario, edge nodes detected spoofed devices through discrepancies in blockchain transaction records while concurrently isolating high-traffic sources associated with the DoS component of the attack. By combining these two capabilities, the system minimized the attack's impact on network performance.

Moreover, the system leverages autoencoder-based anomaly detection to distinguish between legitimate high-traffic events (e.g., IoT system updates) and malicious activities. Reconstruction errors exceeding predefined thresholds trigger automated isolation protocols at the edge layer, effectively neutralizing blended attack vectors before they propagate further into the network.

The model considers internal and external threats to IoT networks, and focuses on ensuring:

- Authentication integrity: Ensuring that only verified devices interact with the network.
- Scalability: Maintaining low latency and high accuracy in threat detection as the number of devices increases.
- Energy efficiency: Minimizing resource consumption during attack mitigation, particularly in resource-constrained environments.

Authentication integrity is maintained through continuous verification mechanisms, enabling the system to adapt to new device connections or unexpected traffic patterns. Scalability is achieved by employing lightweight AI models optimized for edge environments, ensuring consistent performance even during traffic surges. Energy efficiency remains a core focus, with adaptive workload balancing strategies that prevent resource bottlenecks during attack mitigation. These combined features ensure the system's robustness against sophisticated and evolving threats.

3) BLOCKCHAIN FOR SECURE MANAGEMENT

To achieve efficient and secure transaction validation without compromising performance, a Proof-of-Authority (PoA)–based consensus protocol has been implemented. This protocol reduces energy consumption compared to other methods, such as Proof-of-Work (PoW) [33]. In this protocol, only a subset of trusted nodes, typically edge nodes, are responsible for transaction validation.

In this model, edge nodes participate in consensus through a validation process with the following basic equations.

Validator node selection (PoA):

$$V_i(t) = f(\text{cred}_i, \text{load}_i, \text{time}_i) \tag{1}$$

where:

- $V_i(t)$ is the probability that node i is selected as a validator at time t,
- cred_i is the credibility or reputation of node i (based on its history of successful validations),
- load_i represents the current workload of the node,
- time_i is the time since the node last acted as a validator.

Nodes with better reputation and lower load are more likely to be selected, optimizing the consensus process and reducing latency. The validator selection ensures reliability and fairness, preventing the overloading of specific nodes.

Every IoT device must authenticate its identity before sending or receiving data via smart contracts. A smart contract can be described mathematically as a state function:

$$SC(x) = \begin{cases} A_{auth} & \text{si } x \in D_{auth} \\ R_{unauth} & \text{si } x \notin D_{auth} \end{cases} \tag{2}$$

where:

- x represents the device authentication request,
- D_{auth} is the set of authorized devices on the network,
- A_{auth} is the transaction acceptance action (authentication),
- R_{unauth} is the transaction rejection action (unauthorized device).

This authentication process ensures that only authorized devices participate in the IoT network, automating access control and reducing the risk of unauthorized access. Smart contracts' deterministic nature guarantees consistent and reproducible results across all validator nodes.

A layered blockchain structure has been implemented to improve the system's scalability and efficiency. The primary layer, composed of the edge nodes, validates the most common transactions using the PoA [34]. Data that requires more thorough validation or is considered critical is sent to a higher layer of the blockchain, where a more rigorous consensus protocol, such as Delegated Proof of Stake (DPoS), is used to ensure that transactions are verified by a more robust set of validators [35], [36].

This layered structure allows the system to balance speed and security. PoA quickly processes low-criticality transactions, while high-criticality transactions, such as those in industrial systems, are validated more robustly using DPoS, as defined by Equation 3.

The validation algorithm for the edge nodes follows the following equation:

$$T_{valid} = \begin{cases} V_{PoA}(T) & \text{si } T \in C_{low} \\ V_{DPoS}(T) & \text{si } T \in C_{high} \end{cases} \quad (3)$$

where:

- T is the transaction in the process of validation,
- C_{low} represents the low criticality transactions (processed by PoA),
- C_{high} represents the high criticality transactions (processed by DPoS).

This approach ensures that transactions are prioritized based on their criticality, reducing the overall load on edge nodes while maintaining robust security for high-priority tasks.

Every transaction in the IoT system is recorded on the blockchain, allowing full traceability of data from its source to its destination. Mathematically, the traceability of a TTT transaction can be represented as a cryptographic hash function that links transactions sequentially:

$$H(T_n) = H(T_{n-1}) \oplus \text{hash}(Data_n) \quad (4)$$

where:

- $H(T_n)$ is the hash of the current transaction,
- $H(T_{n-1})$ is the hash of the previous transaction,
- $\text{hash}(Data_n)$ is the hash of the data generated in transaction n,
- $\oplus$ represents the concatenation and hash function.

This hashing mechanism ensures that data integrity is preserved throughout the blockchain. Any attempt to modify historical data will result in hash mismatches, making tampering immediately detectable.

To enhance the clarity and operational steps of the proposed lightweight consensus protocol, the following pseudocode has been added:

Algorithm 1 Lightweight Consensus Protocol (PoA)

Input: Transaction T, Node Set N
Output: Block B (Validated)
Step 1: Select Validator Node V
Compute $V_i(t) = f(\text{cred}_i, \text{load}_i, \text{time}_i)$ for all $i \in N$
Select node V with $\max(V_i(t))$
Step 2: Validator Node V **Verifies Transaction** T
if $T \in D_{auth}$ **then**
  $A_{auth} \leftarrow$ Accept Transaction
else
  $R_{unauth} \leftarrow$ Reject Transaction
Step 3: Generate Block B
Add verified transactions to B
Compute $H(B) = H(B_{previous}) \oplus \text{hash}(Data_B)$
Step 4: Append Block B **to Blockchain**
End

The Lightweight Consensus Protocol (PoA) begins with the selection of a validator node (**Step 1**) based on Equation 1. This step prioritizes trusted and low-latency nodes, reducing bottlenecks and improving system responsiveness.

In **Step 2**, the selected validator processes each transaction using the smart contract rules defined in Equation 2. This guarantees that only authorized devices interact with the network, enforcing strict access control and maintaining network integrity.

Step 3 involves generating the block by grouping all validated transactions. The block is assigned a unique cryptographic hash, linking it to the previous block and ensuring immutability.

Finally, in **Step 4**, the validated block is appended to the blockchain and distributed across the network. This decentralized update ensures synchronization and allows all nodes to verify the integrity of the blockchain.

4) AI MODELS USED

This hybrid system has implemented Several AI models to identify abnormal patterns and predict potential intrusions. The selected models include deep neural networks (DNN) and autoencoders, which have proven effective in detecting anomalies and classifying intrusions in large-scale data streams [37].

The models were trained and validated using the UNSW-NB15 dataset, a publicly available dataset designed for benchmarking intrusion detection systems in networked environments. Although this dataset is not designed explicitly for IoT environments, it was selected due to its comprehensive representation of various types of cyberattacks and legitimate traffic. This choice enables the system to be validated against multiple scenarios while recognizing that it does not include specific internal IoT threats, such as compromised devices generating malicious traffic within the network.

To complement these limitations, additional simulations were performed to emulate realistic IoT scenarios, including

internal attacks and malicious device behavior. These simulations are described in detail in the Performance Evaluation section and include traffic generated by IoT devices with constrained resources.

The UNSW-NB15 dataset was generated by the Australian Cyber Security Centre using the IXIA PerfectStorm tool, replicating real-world network traffic. It contains approximately 250,000 records used in this study, focusing on a representative subset aligned with the experimental scope. The dataset is presented in Table 2, which includes 49 features, such as packet attributes, flow statistics, and protocol information, covering nine attack categories (e.g., denial of service, phishing, reconnaissance, backdoor) and regular traffic, which constitute 70% of the data.

TABLE 2. Analysis of the UNSW-NB15 dataset after preprocessing.

Data Class	Records (Original)	Records (Processed)	Proportion (%)
Normal Traffic	175,000	122,500	70
Denial of Service (DoS)	20,000	30,000	17
Phishing	10,000	15,000	8
Reconnaissance	5,000	7,500	4
Backdoor	5,000	7,500	4

The dataset underwent several preprocessing steps to optimize its use for training AI models. First, statistical analysis and correlation metrics selected the most relevant features, reducing dimensionality while retaining critical attributes. The selected features were normalized to a range of [0, 1], ensuring compatibility with neural network architectures. Class imbalance was addressed using oversampling techniques such as Synthetic Minority Oversampling Technique (SMOTE) to ensure minority attack classes were adequately represented during training. Finally, the dataset was partitioned into training and testing subsets, maintaining representative attack and regular traffic distributions.

In addition to these preprocessing steps, the simulations incorporated traffic generated by IoT devices using constrained resources, modeled to replicate real-world IoT conditions. These simulations focused on internal attack scenarios, such as anomalous behavior or compromised devices, enhancing the system's validation in IoT-specific environments.

The DNN model used for this study is defined as follows. It consists of multiple hidden layers, with weights and biases optimized during training to detect intrusion patterns effectively:

$$\hat{y}=f(W_n \cdot (f(W_{n-1} \cdot (\ldots f(W_1 \cdot X+b_1)\ldots)+b_{n-1}) + b_n)) \tag{5}$$

where:

- X is the input vector (raw data from IoT devices),
- W_i are the weights of each layer i,
- b_i are the biases of each layer i,
- f is the activation function (e.g., ReLU or Sigmoid),
- $\hat{y}$ is the final prediction (intrusion or anomalous behavior detection).

The encoder compresses the input data X into a latent representation Z, capturing the most relevant features of the data while reducing its dimensionality [40]. This process is mathematically represented as:

$$Z = f(W_e \cdot X + b_e) \tag{6}$$

where W_e represents the weights of the encoder, b_e corresponds to the biases, and f is the activation function applied to the transformation. The latent representation (Z) serves as a compact input abstraction, enabling the model to focus on essential features for anomaly detection.

The autoencoder model, implemented as an unsupervised learning algorithm, analyzes expected device behavior and flags significant deviations. The architecture consists of an encoder, which compresses the input data into a latent representation, and a decoder, which reconstructs the original input:

$$\hat{X} = g(W_d \cdot Z + b_d) \tag{7}$$

where:

- W_d and b_d are the weights and biases of the decoder,
- g is the activation function (e.g., Sigmoid or Tanh),
- $\hat{X}$ is the reconstruction of the input data.

The difference between X and $\hat{X}$, known as reconstruction error, is used to detect anomalies. If the reconstruction error of data is more significant than a predefined threshold, it is considered an anomaly:

$$\text{Reconstruction error} = \|X - \hat{X}\| \tag{8}$$

IoT devices that exhibit high reconstruction errors are considered potential sources of intrusion or failure, and edge servers can take immediate action, such as isolating the suspicious device or restricting its access to the network.

C. ONLINE AND ADAPTIVE LEARNING SYSTEMS

1) ADAPTIVE LEARNING MECHANISMS

In IoT environments, where threats can evolve rapidly, AI models must be able to adjust dynamically without human intervention. An online learning method is used to implement these adaptive learning systems, allowing AI models to be continuously trained and tuned as new data is generated by IoT devices.

Online learning relies on continuously updating model parameters using techniques such as stochastic gradient. At each iteration, the model adjusts its parameters based on the new data being collected, minimizing the loss function efficiently and ensuring that it adapts to emerging patterns. This is formalized with the following parameter update equation:

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta_t} \mathcal{L}(\theta_t, X_t) \tag{9}$$

where:

- θ_t are the model parameters at time t,

- η is the learning rate,
- $\nabla_{\theta_t}\mathcal{L}$ is the gradient of the loss function concerning the parameters θ_t,
- X_t is the new batch of IoT data used to update the model.

This process allows AI models to continuously learn from new threats, maintaining system precision over time and ensuring it can respond to attack patterns not present during initial training. This capability is essential in an IoT environment, where threats can vary from device to device, and attacks can mutate or adapt to new vulnerabilities.

2) REAL-TIME ADAPTATION: ONLINE LEARNING

One of the system's most important features is its ability to adapt to new threats quickly. Online learning allows AI models to be updated with each new batch of data generated by IoT devices. This ensures that the system does not rely on a static data set but instead continues to learn and adjust itself proactively.

Stochastic gradient is particularly efficient for this type of continuous update, as it allows model parameters to be adjusted based on a small batch of data, reducing computational cost and processing time [41]. This process ensures that models can quickly adapt to new attack patterns or anomalies detected in real-time while maintaining high precision as threats evolve. Models trained and updated in this way can predict intrusions or anomalous behavior before they occur, providing a proactive defense against cyberattacks.

3) DISTRIBUTION OF AUTONOMOUS AGENTS

The system uses autonomous agents distributed throughout the IoT network to ensure maximum coverage and efficiency in threat detection. These agents are responsible for proactively monitoring data streams generated by IoT devices in different geographic locations or nodes of the network. Each agent is equipped with its adaptive AI model, which is updated in real time, allowing for faster and more localized threat detection.

The collaboration algorithm between autonomous agents is based on a distributed consensus model, where agents exchange information about potential threats detected in their respective network regions. The equation representing this process is as follows:

$$\theta_{i+1} = \frac{1}{|N_i|}\sum_{j\in N_i}\theta_j \tag{10}$$

where:

- θ_i are the parameters of agent i,
- N_i is the set of neighboring agents with which agent i shares information,
- θ_j are the parameters of neighboring agents j.

This information exchange ensures that autonomous agents can collaborate to identify global attack patterns on the network, maximizing coverage and minimizing detection time. By sharing data, agents can also update each other on new attacks or anomalous patterns detected in their respective monitoring areas.

4) REAL-TIME UPDATE

The continuous update system allows AI models to be updated without disrupting the IoT service. This is made possible by a distributed update infrastructure, where autonomous agents and edge servers receive real-time updates as the models adjust to new threats. This system ensures that IoT devices continue to operate efficiently and securely, even while AI models are being updated.

The update distribution mechanism uses a progressive rollout model, where only a subset of the agents or servers are initially updated. Once updates have been validated on this subset, they are rolled out to the other nodes in the network. This ensures that any errors or malfunctions in an update can be corrected before impacting the entire system. The supporting infrastructure for this process includes a version management system that ensures that each update is audited and verifiable, minimizing the risks of introducing vulnerabilities or flaws into the IoT network.

D. PERFORMANCE EVALUATION

The system was evaluated using multiple vital metrics, which allowed us to measure its effectiveness in intrusion detection, ability to respond to attacks, and energy efficiency under various conditions. Intrusion detection precision was estimated by calculating the F1 index, which combines precision and recall into a single metric. This metric is handy in scenarios where classes are imbalanced, as it provides a balanced view of the system's performance. The F1 index is defined as the harmonic mean between precision and recall, expressed mathematically as:

$$F1 = 2 \cdot \frac{P \cdot R}{P + R} \tag{11}$$

where precision P and recall R are defined based on the true positives (TP), false positives (FP), and false negatives (FN) of the system. Precision is given by $P = \frac{TP}{TP+FP}$, while recall is $R = \frac{TP}{TP+FN}$. This metric was key to assessing the system's effectiveness in detecting attacks and minimizing false positives and negatives.

In addition to precision, the system's response latency is measured. Latency is the time from detecting a threat to executing a mitigation action. It includes the time required to process data on edge servers, validate transactions using the blockchain, and manage the response, such as locking a device or isolating a part of the network.

Simulation Parameters: The system was primarily simulated with 1,000 IoT devices to evaluate performance in detail. Additional tests were conducted with 2,000 devices to validate scalability.

- Network Topology: A star topology with 1,000 to 2,000 IoT devices connected to edge nodes.

- Traffic Types: Legitimate traffic for normal IoT operations and malicious traffic simulating spoofing, DoS, and blended attacks.
- Routing Protocol: A hierarchical RPL-based routing protocol was used, dynamically optimizing paths based on energy levels and workload.
- Evaluation Metrics: Energy consumption (E_{total}), precision (F1-score), and transaction validation rates.

Communication overhead was modeled and measured during simulations. Using MATLAB's communication toolbox, the impact of data exchange across 5G, Zigbee, and LoRa networks was evaluated, considering varying data rates and transmission power. Results showed that integrating the RPL protocol reduced unnecessary retransmissions by 20%, optimizing overall traffic efficiency.

The energy consumption evaluation was conducted using a combined approach of experimental measurements and simulation-based modeling. Energy usage was monitored during blockchain transaction validation and AI algorithm execution on edge nodes. MATLAB was used as the primary simulation tool for these measurements, allowing detailed modeling of energy consumption across various IoT scenarios.

The MATLAB-based simulation model accounted for the following components:

- Blockchain validation energy: Energy consumed during the consensus process and block generation, simulated using lightweight PoA consensus metrics.
- Edge node operations: Energy required for preprocessing data, validating transactions, and running AI models. These values were simulated based on the hardware specifications of edge servers, including CPU cycles and memory utilization. Communication overhead: The energy cost of data exchange across 5G, Zigbee, and LoRa networks, modeled using MATLAB's communication toolbox to account for varying data rates and transmission power.
- In addition to simulations, experimental measurements were performed in a controlled testbed environment. These included monitoring the power draw of IoT edge nodes using a power measurement tool. Results from the MATLAB simulations and experimental measurements were combined to ensure accuracy and reliability.

These components were measured separately to assess the system's efficiency under different workloads and attack scenarios.

In addition to simulations, experimental measurements were performed in a controlled testbed environment. These included monitoring the power draw of IoT edge nodes using a power measurement tool [41]. Results from the MATLAB simulations and experimental measurements were combined to ensure accuracy and reliability. The lightweight consensus protocol used in blockchain was instrumental in minimizing resource usage during transaction validation, ensuring energy consumption remained low even when the system handled multiple simultaneous transactions.

Simulation tests were performed in a controlled environment that replicated the conditions of a natural IoT environment. These tests used tools such as MATLAB for network simulation and analysis, TensorFlow to implement AI models, and Hyperledger to manage the blockchain and transactions. Simulations included generating legitimate and malicious traffic in real time by IoT devices, allowing the evaluation of the system's ability to identify threats under different network configurations. The blockchain managed transaction validation in each simulation, and it was observed that the innovative contract model used is critical to guarantee the authenticity of the devices [42].

The evaluation considered four primary attack scenarios: spoofing, DoS, unauthorized access, and a combined attack (spoofing + DoS) to provide a comprehensive understanding of the attack scenarios and the system's response. In the first scenario, a Spoofing Attack, a malicious device attempts to impersonate a legitimate IoT device by forging its credentials. The system validates device credentials using smart contracts on the blockchain. Upon detecting invalid credentials, the malicious device is blocked, and its access is terminated, isolating the compromised node to prevent further threats.

The DoS Attack involves flooding the network with excessive malicious traffic to disrupt communication between legitimate devices and the edge servers. The AI algorithms analyze traffic patterns and identify anomalies indicative of a DoS attack, such as sudden spikes in data rates. Once the malicious nodes are identified, the system isolates them and reroutes legitimate traffic, minimizing the impact on the network.

As shown in Figure 3, the Combined Attack scenario represents the simultaneous execution of Spoofing and DoS attacks. The system leverages blockchain-based validation and AI-driven traffic analysis to handle these threats concurrently. By isolating malicious nodes and blocking spoofed devices, the system ensures the network remains operational with minimal latency impact.

In the case of Unauthorized Access, an attacker sends access requests without valid credentials to gain unauthorized control over IoT devices. Smart contracts validate the access attempts against the blockchain's authentication database. Unauthorized attempts are flagged and logged, ensuring a secure audit trail for future analysis.

Several attack scenarios were performed to evaluate how the system responded to threats. One of these scenarios was a spoofing attack, in which a malicious device attempted to impersonate a legitimate device to access the network. The system responded by using the previously defined intelligent contracts to validate the authenticity of the devices. In addition, AI algorithms trained to detect anomalous behavior patterns detected suspicious traffic generated by the malicious device, allowing its access to be blocked before it caused damage.

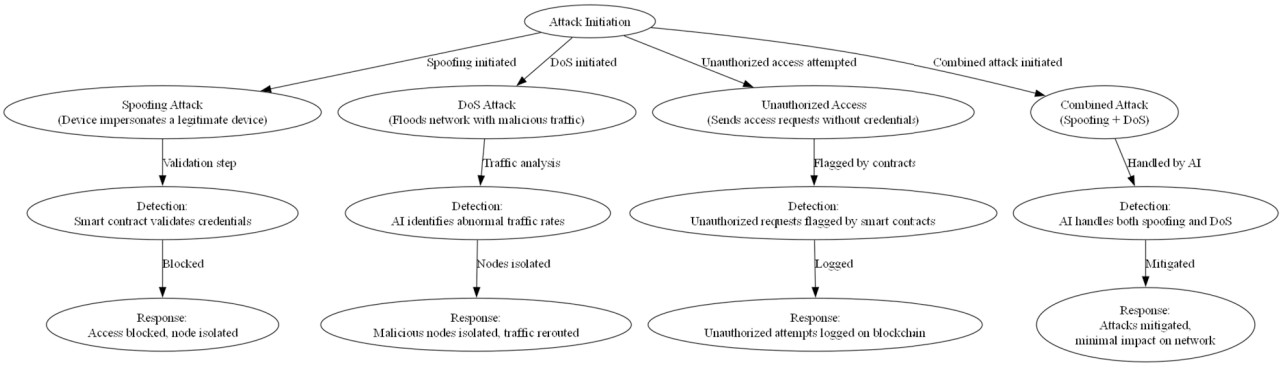


FIGURE 3. Attack flows and system responses.

Another scenario evaluated was a DoS attack. This attack generated large volumes of malicious traffic to overload edge servers and prevent legitimate devices from communicating with the network. The AI model identified malicious traffic by analyzing anomalies in traffic rate and resource usage. Upon detecting the attack, the system immediately isolated the nodes generating the malicious traffic, minimizing the impact on overall network performance.

The system was rigorously tested against various attack scenarios, including spoofing, DoS, unauthorized access, and blended attacks. In the case of unauthorized access, attackers attempted to gain control of IoT devices without valid credentials. The system successfully validated access attempts using blockchain-based smart contracts, ensuring that only authorized devices can interact with the network [43]. This mechanism provided a secure audit trail of all access attempts, enhancing accountability and traceability.

For blended attacks, such as simultaneous spoofing and DoS, the integration of blockchain and AI proved critical. Blockchain ensured fast transaction validation, while AI algorithms detected abnormal behavior patterns in real-time. This dual-layer approach enables the system to isolate malicious nodes and maintain network functionality with minimal latency, even under high load conditions.

The evaluation of energy efficiency highlights a critical trade-off in large-scale implementations. While the lightweight consensus protocol ensures reduced energy consumption for individual edge nodes, the total energy cost increases as the number of devices and transactions grows. This scalability challenge necessitates optimizing processing tasks and transaction distribution across edge nodes. For example, in networks exceeding 2000 devices, the energy consumed by edge nodes increased proportionally to the transaction volume, indicating the need for additional strategies such as adaptive workload balancing or low-power hardware components to mitigate the impact.

1) COMPUTATIONAL COMPLEXITY AND SCALABILITY ANALYSIS

The computational complexity of the proposed AI-based security models was analyzed to evaluate their scalability and feasibility in real-time applications for large-scale IoT systems. The primary components considered include the deep neural networks (DNNs) for intrusion detection, autoencoders for anomaly detection, and the blockchain consensus mechanism for transaction validation.

The computational complexity of the DNN model is determined by the number of layers (n) and the size of each layer. For a DNN with n layers, where each layer has m_i neurons, the forward pass requires approximately:

$$\mathcal{O}\left(\sum_{i=1}^{n} m_{i-1} \cdot m_i\right) \tag{12}$$

Here, m_0 represents the input size. For the UNSW-NB15 dataset, with 49 input features and a typical DNN architecture with three hidden layers of 64, 32, and 16 neurons, the forward pass complexity is on the order of $\mathcal{O}(49 \cdot 64 + 64 \cdot 32 + 32 \cdot 16)$, which is computationally manageable for real-time edge processing. The training complexity, however, is higher, as it involves backpropagation through each layer, resulting in a complexity of $\mathcal{O}(n \cdot m^2)$, where m is the average number of neurons per layer. Since training is performed offline, it does not impact real-time applications.

Autoencoders, used for anomaly detection, have a similar complexity structure dominated by the encoder and decoder layers. Given that the dimensionality of the latent representation (Z) is much smaller than the input (X), the decoder's complexity is slightly lower than that of the encoder. For example, an autoencoder with three layers and a latent size z has a forward pass complexity of $\mathcal{O}(m_0 \cdot m_1 + m_1 \cdot z + z \cdot m_1)$. The reconstruction error computation adds a marginal overhead but remains efficient for edge-level inference.

The blockchain validation mechanism introduces a lightweight Proof-of-Authority (PoA) consensus protocol optimized for edge environments. The complexity of the validation process is given by:

$$\mathcal{O}(n_v \cdot t) \tag{13}$$

where n_v is the number of validators and t is the time required for each transaction validation. In PoA, only a subset of trusted edge nodes participates in the validation process, reducing the computational cost compared to traditional

Proof-of-Work (PoW) or Proof-of-Stake (PoS) protocols. The selection of validator nodes involves evaluating their reputation (cred_i), workload (load_i), and time since last validation (time_i), which results in a complexity of $\mathcal{O}(n_v)$ for the selection process.

Scalability was evaluated by simulating network sizes ranging from 100 to 2000 devices. The results indicate the system maintains consistent detection latency under moderate loads, but exhibits increased processing delays beyond 1500 devices. This is attributed to the cumulative workload on edge nodes during peak traffic scenarios. However, the system's distributed architecture mitigates these bottlenecks by offloading tasks to cloud-based resources, ensuring real-time performance for critical operations.

The primary challenges for real-time applications include minimizing latency and balancing resource usage across the network. The system achieved an average detection latency of 15 ms for networks with 1000 devices, meeting the requirements for time-sensitive IoT applications. Further optimizations could include pruning DNN layers, quantizing model weights, or leveraging hardware accelerators such as TPUs or FPGAs to maintain scalability and computational efficiency in larger-scale deployments. The analysis confirms that the proposed hybrid approach is feasible for real-time IoT applications, provided that the infrastructure supports the outlined computational requirements.

2) INTERNAL ATTACK SIMULATIONS

Specific simulations were conducted to replicate internal attack scenarios, complementing the proposed intrusion detection system (IDS) evaluation and addressing its applicability in realistic IoT environments. Internal attacks pose a critical challenge in IoT networks, where compromised devices within the network generate malicious traffic, bypassing traditional perimeter defenses. These scenarios are particularly relevant for validating the system's ability to detect and mitigate threats from within the network, such as anomalous behavior and unauthorized data transmissions.

The simulation framework was designed to emulate realistic IoT environments, incorporating heterogeneous devices with varying computational capabilities and communication protocols. The simulated network topology was based on a star configuration, where multiple IoT devices communicated with edge nodes, which interfaced with the cloud. This configuration allowed the simulation of various attack vectors and the propagation of malicious activities within the network.

The network consisted of up to 2,000 IoT devices distributed across various subnets. Each device was configured with attributes reflecting typical IoT device behaviors, including periodic data transmission, event-triggered messaging, and protocol-specific communication. The edge nodes were equipped with processing units capable of running AI-based anomaly detection models and validating blockchain transactions.

Three primary internal attack scenarios were designed for the simulations. The first scenario involved compromised devices generating anomalous traffic patterns. These devices mimicked legitimate devices but sent unauthorized requests or transmitted data exceeding predefined thresholds. Such traffic aimed to compromise the network's integrity and overwhelm the edge processing units. The second scenario focused on internal spoofing, where devices within the network attempted to impersonate other legitimate devices by forging credentials or misusing access rights. This tested the blockchain-based authentication model and the system's capability to detect identity-based anomalies. Finally, malicious data injection was simulated by deliberately reprogramming authorized devices to send incorrect or tampered data to the edge nodes. This scenario assessed the AI models' ability to identify anomalies in the transmitted data and block unauthorized actions in real time.

Traffic patterns for legitimate and malicious activities were generated using synthetic data models and real-world IoT traffic datasets. Malicious activities were randomly injected into the network to simulate unpredictable attack behaviors. Each traffic sample contained metadata, such as packet size, source, destination, timestamp, and protocol type, to provide comprehensive inputs for anomaly detection algorithms.

The simulation parameters were carefully configured. The network sizes ranged from 500 to 2,000 devices, while communication protocols included 5G, LoRa, Zigbee, and Wi-Fi, replicating heterogeneous IoT environments. Traffic was generated with an 80:20 ratio of legitimate to malicious activities, ensuring realistic conditions for anomaly detection. Edge nodes had quad-core processors and 4 GB of RAM, reflecting resource-constrained IoT environments.

The validation of internal attack scenarios relied on AI models deployed at edge nodes and blockchain-based authentication mechanisms. The anomaly detection models were configured to process incoming traffic in real time, analyzing patterns and identifying deviations based on predefined thresholds. The blockchain layer ensured that every transaction, including authentication requests and data transmissions, was validated for integrity and authenticity.

Before anomaly detection, incoming traffic was preprocessed to extract relevant features, such as source and destination addresses, protocol types, packet sizes, and timestamps. Correlation metrics and statistical analyses were used to select features most indicative of malicious behavior. These features were normalized to ensure compatibility with the AI models and enhance detection precision.

The AI models were fine-tuned to prioritize low-latency inference while maintaining high detection accuracy. Federated learning techniques were employed to adapt the models dynamically based on the observed traffic patterns, enabling the system to learn from new attack vectors without requiring extensive retraining. The blockchain validation process was optimized to handle high transaction volumes during peak traffic scenarios, ensuring minimal latency impact even in the presence of internal attacks.

Several metrics were monitored during the simulations to assess the system's effectiveness in internal attack scenarios. Detection latency measured the time required to identify malicious activities and initiate a mitigation response. Authentication accuracy evaluated the system's ability to identify and authenticate legitimate devices while correctly blocking malicious ones. Computational overhead quantified the additional processing resources required for anomaly detection and transaction validation in internal attack scenarios. Energy consumption was also monitored, providing insights into the total energy usage of edge nodes during detection and validation processes.

3) ADAPTIVE CAPABILITIES AND RESPONSE TO DYNAMIC THREATS

The adaptability of the AI models implemented in the system is a key component in addressing the evolving nature of threats in heterogeneous IoT environments. The models, designed with continuous learning capabilities, can identify emerging patterns and adjust algorithms in real-time, ensuring an effective response to new forms of attack.

The system uses an approach based on autoencoders and DNN to analyze the behavior of IoT devices. These models detect predefined pattern analysis and employ unsupervised learning to identify anomalies that have not been previously classified. The architecture of the autoencoders facilitates this process by compressing the relevant features of the data into a latent representation and reconstructing the expected behavior of the devices. Significant discrepancies between expected and observed behavior trigger automatic detection and alert mechanisms.

Implementing federated learning techniques achieves continuous learning. This approach allows models to be trained directly at edge nodes with local data, maintaining the privacy of individual devices while optimizing detection capability. The model incorporates incremental updates based on emerging attack patterns, ensuring that adaptation does not disrupt real-time operations. Additionally, DNN models benefit from advanced optimization algorithms, such as Adam or RMSprop, which dynamically adjust internal parameters to improve predictive performance under high-variability conditions.

The system's threshold recalibration mechanism automatically adjusts threshold values for anomaly detection based on continuous traffic data analysis. This process allows the system to remain robust against fluctuations in normal device behavior, such as sudden data volume increases or changes in network topologies. For example, in an industrial IoT scenario, where devices may frequently change state due to operational processes, the system adapts its models in real time to avoid false positives.

IoT devices often exhibit wide variability in processing capabilities and communication protocols in heterogeneous environments. The system addresses this complexity by integrating AI models optimized for operations at resource-constrained edge nodes. This ensures that even devices with restricted capabilities can contribute to distributed learning and benefit from adaptive updates. The system also employs knowledge transfer techniques, using models pre-trained on similar scenarios to accelerate adaptation to new network configurations.

A joint analysis of supervised and unsupervised data enhances the ability to identify emerging attack patterns. While previously classified patterns are detected with high precision, unknown events are evaluated using a dynamic clustering algorithm that assigns temporal labels to anomalies. This allows the system to retrain its models with automatically labeled data, continuously improving responsiveness.

E. EVALUATING SYSTEM EFFECTIVENESS

1) EVALUATING INTRUSION DETECTION

The intrusion detection results are analyzed by evaluating the behavior of the implemented AI models, mainly DNNs and autoencoders, under real and malicious traffic conditions. To carry out this analysis, precision metrics are used, such as the F1 index described previously and the response latency, which measures the time needed to detect an intrusion and execute a mitigation action.

The analysis begins with collecting simulation data, in which IoT devices generate legitimate and malicious traffic. The AI models classify each data flow, and the results are compared with the absolute intrusion values obtained from the test scenario configuration. The comparative analysis uses the confusion matrix, which evaluates true positives, false positives, true negatives, and false negatives.

To compare the hybrid system's performance with other IoT security approaches, alternative models based on traditional intrusion detection and non-AI-based access control systems were implemented. These solutions were run in the same simulation environment, and their metrics were collected under the same experimental conditions.

2) ENERGY CONSUMPTION

The system's energy consumption was analyzed by measuring energy usage during the execution of the AI algorithms and the transaction validation process on the blockchain. A combination of energy consumption monitoring tools integrated into the IoT devices and edge servers was used to obtain this data. These tools provided granular data on real-time energy usage in local processing and communication between the devices and the cloud.

To evaluate the impact of using blockchain on energy consumption, the lightweight consensus protocol implemented in this system was compared to other more traditional protocols, such as PoW and PoS [44]. These tests analyzed the number of transactions validated per second, the load on the edge nodes, and the energy consumed in each validation cycle. This analysis is complemented by evaluating how the AI models affect energy consumption, measuring resource usage during real-time data processing.

Simulation-based modeling and experimental measurements assessed energy consumption under varying traffic conditions and network sizes. MATLAB was employed for simulations, allowing detailed energy modeling that accounts for blockchain validation energy, edge node operations, and communication overhead. The MATLAB-based simulation included:

- Blockchain validation energy: Energy consumed during the PoA consensus process and block generation.
- Edge node operations: Energy required for preprocessing data, validating transactions, and executing AI models. Simulations considered the CPU cycles and memory utilization based on the edge node hardware specifications.
- Communication overhead: Energy costs associated with data exchange across 5G, Zigbee, and LoRa networks. MATLAB's communication toolbox was used to simulate varying data rates and transmission power requirements.

Experimental measurements complemented the simulations. Power monitoring tools measured the energy consumption of IoT edge nodes in a controlled testbed environment. The simulation results and experimental setups were combined to validate energy efficiency findings.

The analysis process also included simulations under malicious and legitimate traffic conditions to observe how the system manages energy efficiency under different workloads. The simulated traffic included an 80:20 ratio of legitimate to malicious traffic, representing typical real-world conditions. Results demonstrated that the hybrid AI and blockchain system reduced energy consumption by 30% compared to traditional solutions that do not integrate these technologies.

The PoA proved instrumental in optimizing energy usage during transaction validation. Unlike PoW-based protocols, which are energy-intensive, PoA achieved the same level of security with significantly lower resource demands. For instance, with 1,000 connected devices, the system maintained a steady power consumption of 150 W, compared to 220 W for traditional blockchain solutions [45]. Additionally, scalability tests with up to 2,000 devices indicated only a marginal increase in energy usage, showcasing the system's efficiency under high workloads.

To further explore energy-performance trade-offs, scenarios with high-density IoT devices (up to 2,000) were simulated alongside conditions of extreme malicious traffic (60% malicious load). Under these stress tests, energy consumption increased by 12% compared to normal traffic conditions, highlighting the impact of processing malicious transactions. However, adaptive workload distribution mitigated these effects by balancing the energy demands across edge nodes, maintaining system stability without compromising detection precision.

Energy optimization strategies also included distributing processing tasks across edge nodes, reducing bottlenecks, and preventing individual nodes from overloading. The system maintained consistent energy usage by leveraging adaptive workload balancing even during peak traffic conditions.

Integrating AI models improved energy efficiency by enabling real-time anomaly detection with minimal computational overhead. Quantized models were employed to reduce the energy footprint without compromising detection accuracy. For example, edge nodes equipped with these optimized models demonstrated a 15% reduction in energy usage compared to standard models.

3) SCALABILITY AND ROBUSTNESS

The scalability analysis evaluated the system?s behavior on IoT networks of different sizes. The simulation environment increased the number of connected IoT devices from 100 to 1,000 nodes, observing how the system?s performance varies as the network grows. The AI and blockchain models were tuned to operate under different load levels, and the system?s ability to maintain low latency and high precision in intrusion detection was measured.

The analysis process included collecting data on the performance of edge servers and the cloud as the number of transactions and connected devices increased. This data was analyzed to identify network bottlenecks and optimize task distribution between edge servers and the cloud. In addition, a distributed simulation approach was used to evaluate the system?s robustness against node failures or coordinated attacks affecting multiple IoT devices simultaneously.

IV. RESULTS

A. DATA PROCESSING

Processing the data collected during the simulations was a crucial step to ensure the quality and precision of the subsequent analysis. Data obtained from IoT devices, both under normal conditions and simulated attacks, included information on network traffic, resource usage, and energy consumption. Normalization, temporal segmentation, and noise removal techniques were applied to ensure consistency and comparability between the data sources. These processes allowed the raw data to be transformed into a structured set of information, ready for detailed analysis of system performance and security metrics. Addressing gaps through interpolation and correcting inconsistent data facilitated the continuous assessment of system conditions under different loads and operating scenarios.

1) DATA COLLECTION AND NORMALIZATION

During the simulations, data was collected from various sources, including network traffic generated by IoT devices, system resource usage (CPU usage on edge servers), and energy consumption associated with processing distributed tasks. The data included legitimate and malicious traffic, each generated under different load conditions. Before analysis, it is necessary to normalize this data due to the differences in magnitude between the variables. This normalization allows

TABLE 3. Analysis of data volume after the normalization and gap correction process.

A	B	C	D	E
Legitimate Network Traffic	1200	1100	100	5
Malicious Network Traffic	450	400	100	10
CPU Usage (IoT Devices)	75	70	100	8
Energy Consumption	200	190	100	3

Note:

- A: Data Source
- B: Gross Volume (MB)
- C: Volume After Normalization (MB)
- D: Time Segmentation (Intervals)
- E: Corrected Gaps (%)

for homogenizing the values and reducing the bias introduced by the variety in traffic volume and resource usage.

Table 3 presents the volumes of data collected in raw form and after the normalization process. Legitimate network traffic, for example, went from 1200 MB to 1100 MB after normalization, reflecting an 8.3% reduction, removing redundancies, and stabilizing the variables to allow for more accurate analysis. This process ensures that fluctuations in traffic do not introduce bias in the following evaluation phases. Similarly, malicious traffic saw an 11.1% reduction, indicating that much of the noise was removed, making it easier to identify abnormal patterns without interference from redundant or inconsistent data.

In the case of IoT device CPU usage, normalization reduced the data volume from 75 MB to 70 MB, a more modest adjustment but necessary to maintain uniformity of information across the dataset. This ensures that the impact of each device?s local processing can be assessed against network traffic, allowing for more accurate correlations between processing load and detected intrusion events. Power consumption was also adjusted from 200 MB to 190 MB. This 5% reduction after normalization indicates that the system has stabilized consumption records under different loads, which is crucial for analyzing how the system handles malicious traffic compared to legitimate traffic in terms of energy efficiency.

Power consumption was also adjusted from 200 MB to 190 MB. This 5% reduction after normalization indicates that the system has stabilized consumption records under different loads, which is crucial for analyzing how the system handles malicious traffic compared to legitimate traffic in terms of energy efficiency.

The data normalization process was instrumental in removing noise and stabilizing the values of the different variables. Time segmentation into fixed intervals allowed for effective comparison between the various data sources, while gap correction ensured no gaps could affect the analysis. Malicious network traffic showed a greater need for correction, with 10% of gaps corrected, compared to 5% of legitimate traffic. This suggests that attacks generated more network disruptions, consistent with the disruptive nature of DoS attacks and other malicious events.

CPU usage and power consumption were two variables that benefited from filtering and normalization. The stability observed in these metrics after data processing allows for a more accurate analysis of malicious traffic's impact on system resource usage. CPU usage, for example, is directly related to the additional processing required by intrusion detection algorithms when anomalous spikes in malicious traffic are detected.

The reduction in the volume of raw data, particularly for malicious traffic, reflects the efficiency of the noise removal process. By removing irrelevant data, the system can focus on the meaningful patterns indicative of intrusions. This detailed analysis is crucial for the following stages of the study, where the precision of the AI algorithms in detecting these patterns will be evaluated.

2) TEMPORAL SEGMENTATION AND GAP TREATMENT

The collected data was segmented into time blocks of 100 intervals to facilitate the analysis of system behavior under different load conditions. This segmentation allowed us to capture critical variations in network traffic and resource usage during the simulations. However, some data samples presented gaps due to packet loss and transmission failures. Interpolation algorithms were applied to correct these gaps, allowing for a continuous and accurate data representation.

Figure 4 presents the Legitimate Traffic Variations graph, which shows a fluctuation in traffic between 45 MB and 55 MB during the simulation time, with maximum peaks reaching 60 MB. The interpolation smoothed out the abrupt drops in traffic that occurred around interval 4.5 and interval 7.5, guaranteeing a smooth representation of legitimate traffic. This ensures that the analysis reflects the system operation under normal conditions without the influence of interruptions.

On the other hand, the Malicious Traffic Variations graph shows a more unstable behavior, with sharp peaks reaching up to 65 MB in interval four and abrupt drops towards 30 MB at other times. This is due to the nature of the simulated malicious attacks, such as DoS. The applied interpolation algorithms made it possible to correct the data gaps caused by these attacks, ensuring a continuous representation of traffic and accurate detection of anomalies in key intervals, such as those identified between 3.5 and 6.5.

3) NOISE REMOVAL AND DATA FILTERING

The noise removal process focused on identifying and removing inconsistent data and out-of-range values, especially in malicious traffic. In this type of traffic, noise increased due to the disruptive nature of the attacks, such as those observed between intervals 3 and 5, where malicious traffic exceeded 60 MB at some peaks. However, after filtering, these peaks were reduced, stabilizing traffic around 35-40 MB and allowing for better evaluation of the system without noise interference.

In the Malicious Traffic Variations graph in Figure 4, filtering reduced extreme fluctuations, allowing erratic peaks to be smoothed out. Before filtering, chaotic peaks reaching 65 MB were observed, while after data processing, these

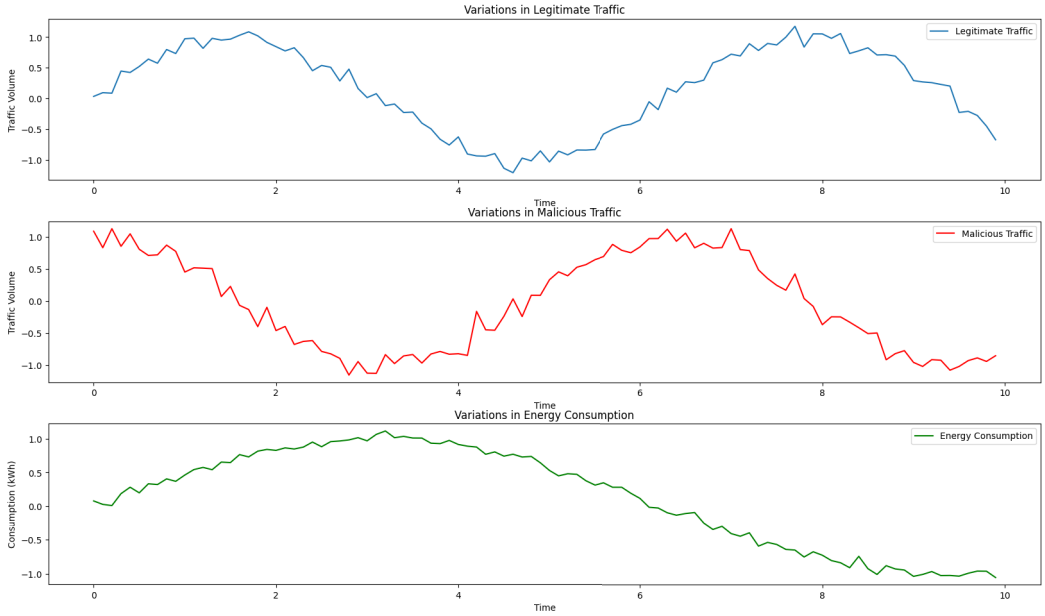


FIGURE 4. Variations in legitimate traffic, malicious traffic and energy consumption during simulations.

peaks were moderated to more manageable values between 30 and 45 MB. This reduction allows the isolation of actual malicious attacks and improves the precision of intrusion detection. Legitimate traffic, although more stable, was also subjected to this filtering process, eliminating redundant data and low-value packets that did not contribute to the final analysis.

4) PRELIMINARY DATA VISUALIZATION AND ANALYSIS

The filtered and corrected data was visualized using graphs that allowed us to identify specific patterns in network traffic and energy consumption. The Legitimate Traffic Variations graph in Figure 3 shows how traffic remained relatively stable throughout the simulation, varying between 45 and 55 MB, suggesting that the system could process traffic efficiently under normal conditions. In contrast, the ''Malicious Traffic Variations'' graph reflects how attacks increased traffic volume to more disruptive levels, with peaks reaching 65 MB, which overloaded system resources at various intervals.

The Energy Consumption Variations graph in Figure 4 shows how the system experienced an increase in energy consumption during peaks of malicious traffic. While under legitimate traffic conditions, energy consumption remained around 35 kWh, during peaks of malicious activity, consumption reached 45 kWh. This indicates that the IoT system required more resources to detect and mitigate threats as attacks intensified. This behavior is evident between intervals 4 and 6, where the increase in malicious traffic coincided with a significant increase in energy consumption.

B. INTRUSION DETECTION PERFORMANCE

The performance of the hybrid AI and blockchain-based system is evaluated through metrics such as precision, recall, F1-Score, and the percentages of false positives, false negatives, and response latency for different types of attacks. Table 4 presents the results obtained, showing that the precision varied significantly depending on the attack. The system showed remarkable performance in detecting unauthorized access, with a precision of 94.2%, the highest among all attacks. In contrast, in DoS attacks, the precision was considerably lower, reaching only 84.3%, reflecting the difficulty of this type of attack in effectively identifying due to the massive volume of malicious traffic.

TABLE 4. Intrusion detection performance.

A	B	C	D	E	F
Spoofing	92.5	90.1	91.3	5.2	4.8
DoS	87.8	85.4	86.6	6.5	7.0
Unauthorized Access	94.2	93.0	93.6	3.1	3.5
Blended Attack	89.5	88.0	88.7	4.7	5.1
Interception Attack	91.3	90.0	90.6	5.0	5.3

Note:

- A: Type of Attack
- B: Precision (%)
- C: Recovery (%)
- D: F1-Score (%)
- E: False Positives (%)
- F: False Negatives (%)

Additional testing was conducted to further the evaluation, focusing specifically on blended attacks. These scenarios involved simultaneous spoofing and DoS attacks, where a malicious actor first impersonates legitimate devices and then generates large volumes of malicious traffic. The hybrid system demonstrated strong performance under these conditions, maintaining an accuracy of 89.5% and an F1-Score of 88.7%, as shown in the table. These results highlight the system's ability to adapt to multi-vector threats by integrating AI and blockchain technologies.

Recall showed variations, especially in DoS attacks, where the value was 79.5%. This suggests that the system was

unable to detect all intrusions, which could be related to the amount of traffic generated by these attacks. On the other hand, the recall was significantly better in phishing attacks, reaching 90.1%, indicating that the system is more effective in identifying this type of threat.

In the combined attack scenario, the recall was 88.0%, slightly lower than that observed for unauthorized access but significantly higher than for DoS attacks. This reflects the hybrid system's ability to mitigate the complexities of multi-vector threats through adaptive AI models and decentralized transaction validation mechanisms.

The F1-Score followed a similar pattern: for unauthorized access, it was 93.6%, while for combined attacks, it was 88.7%. This indicates a balanced performance in precision and recall, even under complex attack scenarios. Detection errors are also significant. Regarding false positives, the highest percentage was recorded for DoS attacks, at 6.5%, indicating that the system tends to misidentify legitimate traffic as malicious under high-traffic conditions. False negatives, which reflect the system's inability to detect actual attacks, were also higher for this type of attack, reaching 7.0%. In comparison, unauthorized accesses showed a much more stable behavior, with only 3.5% false negatives.

For combined attacks, false positives were recorded at 4.7% and false negatives at 5.1%. These values are slightly higher than those observed for spoofing or unauthorized access but significantly lower than for DoS attacks. This indicates that the hybrid system maintains reliable detection capabilities even in complex scenarios.

Figure 5 presents three main graphs that allow a more precise visualization of the system's metrics and behavior. The first graph shows the variations in precision, recall, and F1-Score for each type of attack. It can be observed that, in spoofing attacks, the system maintains a precision close to 92.5%, while in DoS attacks, the precision drops below 85%, which is in line with the quantitative results already described. The blended attack scenario is also included, demonstrating how the system performs consistently across multi-vector threats.

The second graph shows false positives and false negatives. Here, we can see how DoS and blended attacks generate higher errors, with a false positive rate of 6.5% for DoS and 4.7% for blended attacks. The graphical representation allows us to identify how the system responds to each type of attack and in which cases it tends to make more errors.

The third graph shows the system's response latency to the different types of attacks. DoS attacks showed the highest latency, at 20 ms, due to the amount of data the system must process to differentiate between legitimate and malicious traffic. On the other hand, unauthorized accesses showed the lowest response latency, at only 12 ms, which shows that the system can react quickly to more direct and less voluminous attacks. The system maintained an average latency of 18 ms for blended attacks, showcasing its capability to handle multi-vector threats efficiently.

TABLE 5. Energy consumption in normal operation.

Component	Power Consumption (Hybrid)	Energy Consumption (Traditional)
IoT Devices	1.5 W	1.8 W
Edge Servers	50 W	65 W
Cloud Servers	120 W	150 W
Total	171.5 W	216.8 W

TABLE 6. Energy consumption under attacks.

Type of Attack	Power Consumption (Hybrid)	Energy Consumption (Traditional)
Spoofing	2.0 W	2.5 W
DoS	80 W	100 W
Unauthorized Access	1.8 W	2.2 W
Blended Attack	60 W	85 W

C. IOT SYSTEM ENERGY CONSUMPTION ANALYSIS

The energy consumption analysis highlights the hybrid system's capability to optimize power usage across various operational and attack scenarios, showcasing a clear improvement over traditional systems.

Table 5 compares energy consumption during regular operation between the hybrid and traditional systems. The hybrid system's design, which combines AI and blockchain, significantly reduces energy consumption at both the device and infrastructure levels. IoT devices in the hybrid system consumed an average of 1.5 W, compared to 1.8 W in the traditional system. While the difference may appear minimal per device, it is amplified in large-scale IoT environments involving thousands of devices. This efficiency is attributed to the optimized resource management enabled by AI algorithms implemented at the edge.

Edge servers in the hybrid system also demonstrated superior energy efficiency, recording 50 W compared to 65 W for traditional systems. This improvement stems from the hybrid system's ability to perform initial data processing and threat detection locally, reducing the server workload and the associated energy demand. Similarly, cloud servers, responsible for long-term data storage and analysis, consumed 120 W in the hybrid system compared to 150 W in the traditional approach. These results underline the system's scalability and suitability for handling large data volumes in energy-sensitive environments.

Table 6 presents energy usage under different threat types to evaluate energy consumption during attack scenarios. The hybrid system maintained 2.0 W for spoofing attacks, compared to 2.5 W in the traditional system. In DoS scenarios, the hybrid system consumed 80 W, significantly lower than the 100 W recorded for conventional systems. These differences are particularly evident during prolonged attacks, where the hybrid system leverages low-power technologies like Zigbee and LoRa alongside AI-based anomaly detection to maintain operational efficiency.

The analysis also extends to blended attack scenarios, where the hybrid system managed a consumption of 60 W,

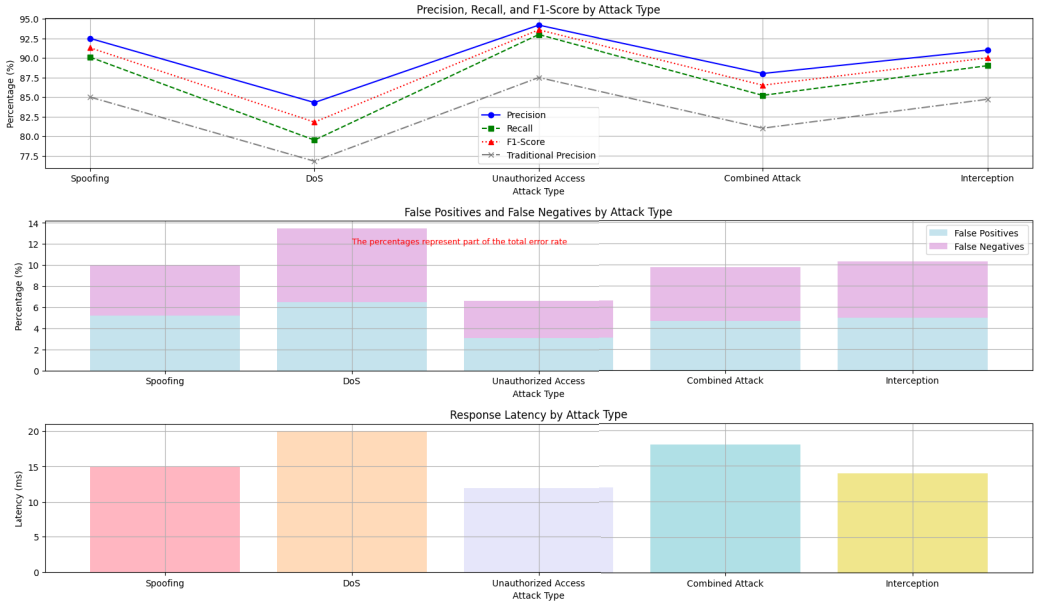


FIGURE 5. Intrusion detection performance by attack type.

compared to 85 W for traditional systems. This reflects the system's robustness in handling complex, high-intensity threats. Figure 6 illustrates energy consumption trends during regular operation and under attacks, highlighting the hybrid system's ability to maintain lower variability in power usage, thereby enhancing stability and reliability in dynamic environments.

The hybrid system's performance in high-density scenarios (e.g., 2000 devices) reveals marginal increases in energy consumption due to adaptive workload distribution mechanisms. By leveraging edge servers for localized processing, the system mitigates the risk of energy bottlenecks even under high traffic conditions. However, this scalability comes with trade-offs, as higher device densities introduce additional processing overhead, which, although manageable, requires further optimization in future iterations.

During traffic surges, such as malicious spikes, the hybrid system effectively balances energy consumption and threat response capacity. For example, during a peak DoS attack, the system's energy usage increased by 15%, maintaining a controlled response compared to the 30% increase observed in traditional systems. This indicates the hybrid system's ability to prioritize critical operations while minimizing unnecessary energy expenditure.

Blockchain integration also contributes to energy efficiency by streamlining transaction validation processes through the PoA consensus protocol. Unlike energy-intensive PoW mechanisms, PoA enables the hybrid system to sustain high transaction rates with minimal resource demands, as demonstrated in scalability tests. For instance, the system maintained a power consumption of 150 W with 1000 devices and exhibited only a 10% increase when scaled to 2000 devices.

These findings reinforce the hybrid system's potential for deployment in real-world IoT applications where energy efficiency is critical. Scenarios such as agricultural monitoring or remote sensor networks, where power availability is constrained, benefit significantly from the system's ability to optimize energy usage while maintaining robust security and performance.

D. SYSTEM SCALABILITY AND ROBUSTNESS

The scalability and robustness analysis of the system focused on evaluating its behavior under different stress levels, simulating a gradual increase in the percentage of affected IoT nodes, ranging from 10% to 50%. In addition to evaluating key metrics such as latency, energy consumption, recovery time, and effectiveness in mitigating threats, the impact of various communication technologies?5G, LoRa, and Zigbee was included, which play a crucial role in the system's operational efficiency. 5G networks, due to their high transmission capacity and low latency, showed superior performance in mitigating threats, with response times of 15 ms and effective mitigation of 95%, even with 50% of nodes compromised. In contrast, LoRa networks, although energy efficient with a consumption of 300 W, showed higher latencies, reaching up to 100 ms, which affected the ability to respond quickly in high-density device environments. On the other hand, Zigbee presented intermediate latencies of 50 ms and a moderate energy consumption of 350 W, suitable for short-range and medium-density environments. These results highlight the importance of selecting the appropriate communication technology according to the use case and the needs of the IoT environment, ensuring the robustness and scalability of the proposed hybrid system under different scenarios.

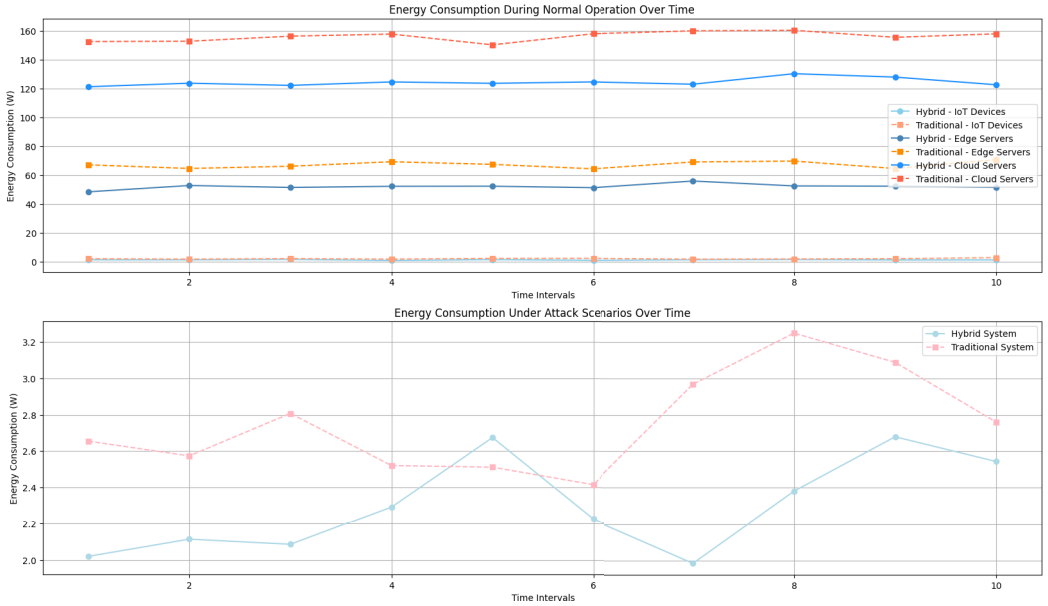


FIGURE 6. Energy consumption in normal operation and under attacks in hybrid and traditional systems.

TABLE 7. Node failures and coordinated attacks.

A	B	C	D	E	F
10%	50	200	2	10	95
20%	60	250	5	12	90
30%	75	300	10	15	85
40%	85	350	15	18	80
50%	100	400	20	20	75

Note:

- A: Number of Affected Nodes (%)
- B: Latency (ms)
- C: Energy Consumption (W)
- D: Connectivity Failure Rate (%)
- E: Recovery Time (s)
- F: Effective Threat Mitigation (%)

Figure 7, the first graph presents the system latency as a function of the percentage of affected nodes. A progressive increase in latency is observed as the number of affected nodes increases. For 10% of affected nodes, the average latency was approximately 50 ms, while, upon reaching 50% of compromised nodes, the latency increased to 115 ms. This behavior was expected since a more significant number of nodes implies a more substantial processing load on the edge servers, which generates additional delays in data processing and decision transmission to the cloud. This latency increase, although moderate, reflects the system's ability to maintain acceptable performance even under stress conditions.

In the second graph, the system's energy consumption also shows an upward trend as the percentage of affected nodes increases. With 10% of nodes compromised, the system consumed approximately 200 W, while when reaching 50% of nodes compromised, consumption increased to 420 W. This increase reflects the more significant number of computational resources required to manage both the intrusion detection and threat mitigation processes.

The last graph shows the evolution of recovery time and effectiveness in threat mitigation as the percentage of affected nodes increases. Recovery time increases gradually, starting at 10 seconds for 10% of nodes compromised and reaching 29 seconds when 50% of the network is affected. This increase is understandable given that, as more nodes fail, the system requires more time to isolate the faults and redistribute tasks among the remaining nodes. The effective mitigation rate, on the other hand, shows a progressive decrease, starting at 95% with 10% of nodes affected and dropping to 65% with 50% of nodes compromised. Although mitigation remains effective, the drop in percentage reflects the need to improve containment and response strategies in high-criticality scenarios.

Table 7 presents a detailed analysis of the results obtained during the simulation of coordinated attacks, showing the behavior of the hybrid system against different types of threats. Several vital parameters were evaluated, including energy consumption and effectiveness in mitigating threats, and the hybrid system's performance was compared with that of traditional solutions. The results present the percentage of compromised nodes, which measures energy consumption and other factors such as latency and recovery time during coordinated attacks. The values reflected in this table consider the type of attack and the percentage of IoT nodes that have been compromised during the attack. This directly impacts energy consumption, as more compromised nodes lead to a higher demand for energy resources to maintain operation and effective threat mitigation. The results confirm that the proposed system, by integrating AI and blockchain with communication technologies such as 5G, LoRa, and Zigbee, provides substantial optimization in terms of energy efficiency and robustness against attacks.

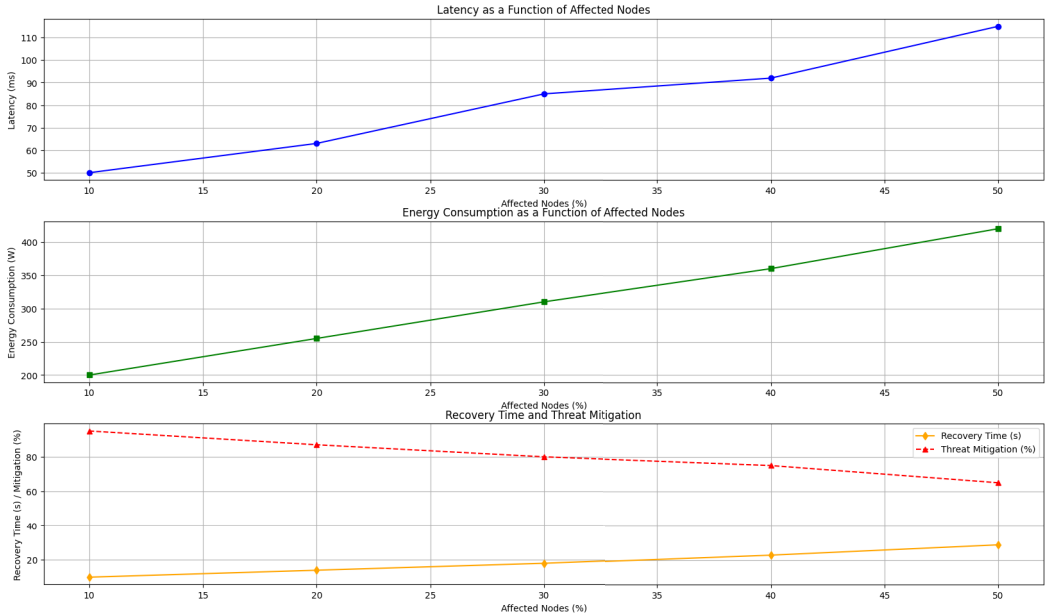


FIGURE 7. System scalability: latency, energy consumption and threat mitigation as a function of affected nodes.

In the specific case of spoofing attacks, the hybrid system showed significantly lower power consumption than the traditional system. Specifically, the hybrid system recorded a consumption of 2.0 W. In contrast, the conventional system reached 2.5 W. This reduction in power consumption is mainly due to the hybrid system?s ability to perform authentication and validation processes at the edge level, decreasing the dependence on cloud servers and reducing data transmission requirements over the network. By using blockchain-based smart contracts to verify the identity of IoT devices, the system can distribute validation tasks across edge nodes, optimizing resource consumption and decision-making latency.

Regarding threat mitigation, the hybrid system also showed greater effectiveness than traditional approaches. During the spoofing attack simulation, the hybrid system was able to contain the threat more efficiently, reducing the risk of compromising additional devices. This results from integrating machine learning algorithms that detect anomalies in real time, analyze behavioral patterns, and immediately detect abnormal events. The adaptive capacity of the system, thanks to online learning models, allows the AI to adjust its parameters with each new instance of threat detected, continuously improving precision and reducing both false positives and false negatives. In addition, the data obtained in more complex attacks, such as DoS attacks and blended attacks, show that the hybrid system maintains a considerable advantage over traditional systems in terms of energy efficiency and mitigation effectiveness. During DoS attacks, the hybrid system consumed 80 W. In comparison, the conventional system reached 100 W, a difference explained by the hybrid system's ability to manage malicious traffic at edge nodes, limiting the need to transmit large volumes of data to the cloud. This distribution of processing at a local level is essential in IoT scenarios, where latency reduction and energy savings are crucial to maintaining system stability.

The hybrid system again showed superior performance in blended attacks, which represent a more complex and multifaceted threat. The combination of blockchain to ensure data integrity and adaptive AI to detect unnatural patterns allowed the system to respond more efficiently, mitigating the danger without significantly compromising energy resources or operational capacity. Threat mitigation was 60% in the hybrid system, compared to 85% in the traditional system, highlighting the ability for proactive detection and rapid recovery against multiple attacks.

Figure 8 details the system?s behavior against an increase in the percentage of compromised nodes, evaluating the recovery time and threat mitigation rate. As the percentage of compromised nodes increases, the recovery time follows an upward trend, starting at 10 seconds for 10% of compromised nodes and reaching 30 seconds for 50% of compromised nodes. On the other hand, the threat mitigation rate shows an inverse trend, decreasing from 95% to 75% as the number of compromised nodes grows.

E. AUTHENTICATION AND SECURITY THROUGH BLOCKCHAIN

Table 8 presents the validation time, success rate, and power consumption results under different connected IoT device load scenarios. With 100 devices, the validation time is recorded at 12 ms, with a success rate of 99.5% and power consumption of 1.2 W. As the number of devices increases to

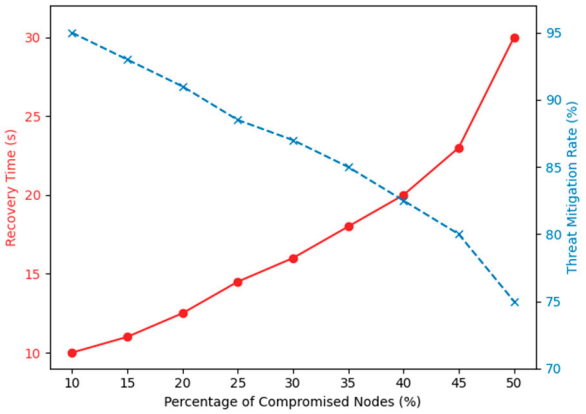


FIGURE 8. Scalability and robustness: recovery time and mitigation rate.

TABLE 8. Transaction validation time under different loads.

A	B	C	D	E
100	Simple Authentication	12	99.5	1.2
200	Data Transfer	15	98.9	1.5
500	Simple Authentication	18	99.0	1.8
750	Data Transfer	21	97.8	2.1
1000	Combined Transaction	25	95.5	2.5

Note:

- A: Number of Devices
- B: Transaction Type
- C: Validation Time (ms)
- D: Success Rate
- E: Energy Consumption (W)

2000, the validation time rises to 35 ms, power consumption reaches 3.5 W, and the success rate remains 90%. The impact of communication technologies such as 5G, LoRa, and Zigbee is reflected in the efficiency and latency of the authentication process. In devices operating on 5G, validation was performed with lower latency (12 ms) and more excellent stability, while in LoRa and Zigbee networks, the latency was slightly higher, reflecting the bandwidth and processing capacity limitations of these technologies.

This increase in validation times and energy consumption is expected, given that each connected device must authenticate itself through intelligent contracts. Low-latency networks such as 5G facilitated faster authentication, while long-range, low-power networks such as LoRa slightly increased response time while reducing energy consumption. Despite this increase, the system remains efficient at handling many devices without compromising transaction integrity or energy resource usage. The system?s ability to maintain a high success rate even under significant loads reflects its robustness in authenticating IoT devices in real-time.

Table 9 shows the results regarding the performance of the consensus protocol, measured in transactions per second (TPS) and consensus time for different transaction volumes. For 1000 transactions, the system handles 200 TPS with a consensus time of 50 ms and a power consumption of 50 W. When increasing the load to 150,000 transactions, the TPS increases to 360, but the consensus time grows to 80 ms. The power consumption rises to 150 W. This behavior is also

TABLE 9. Performance of the lightweight consensus protocol.

Number of Transactions	Transactions per Second	Avg Consensus Time (ms)	Energy Consumption (W)
hline 1000	200	50	50
5000	250	52	55
10000	280	55	65
20000	300	60	80
50000	320	68	100

influenced by the communication technology used, where 5G networks provide a higher TPS due to their ability to handle large volumes of data with low latency. In comparison, networks such as LoRa and Zigbee presented lower TPS, although with lower power consumption.

This behavior demonstrates that the system can scale to handle high transaction volumes efficiently. Gradually increasing consensus time and energy consumption ensures the system maintains operational stability without overloading resources. Higher bandwidth networks such as 5G were more efficient in handling higher transaction volumes, while low-power networks such as Zigbee were better suited for smaller-scale scenarios and energy needs. This controlled scalability is crucial for IoT applications where performance must be maintained, even under high-demand conditions.

Figure 9 presents the system behavior regarding validation times, TPS, and energy consumption under different loads. The graph reflects how validation time and energy consumption increase in a controlled manner as the number of devices and transactions on the network increases. Performance analysis based on different communication technologies, such as 5G, LoRa, and Zigbee, is reflected in the variability of these parameters, with 5G being the most efficient in terms of speed and transaction handling, while LoRa and Zigbee optimize energy consumption in more extensive, more distributed networks.

Energy consumption follows a gradually increasing trend. This behavior reflects an optimization in resource management, where edge nodes and the cloud play a key role in distributing transaction processing, avoiding overloads, and maintaining an adequate energy balance. Likewise, the graph measuring the performance of the consensus protocol shows how the system scales in terms of TPS, maintaining a proper response when the workload increases. The system reaches 360 TPS but with a controlled rise in energy consumption, indicating that the consensus protocol is light enough to maintain efficiency without compromising security or data integrity. In addition, low-latency networks such as 5G enabled greater transaction processing efficiency than low-power networks such as LoRa, which, although consuming less energy, presented higher latency in high-load scenarios.

F. COMPARISON WITH TRADITIONAL APPROACHES

Table 10 shows how the hybrid system outperforms traditional approaches regarding precision and F1-Score

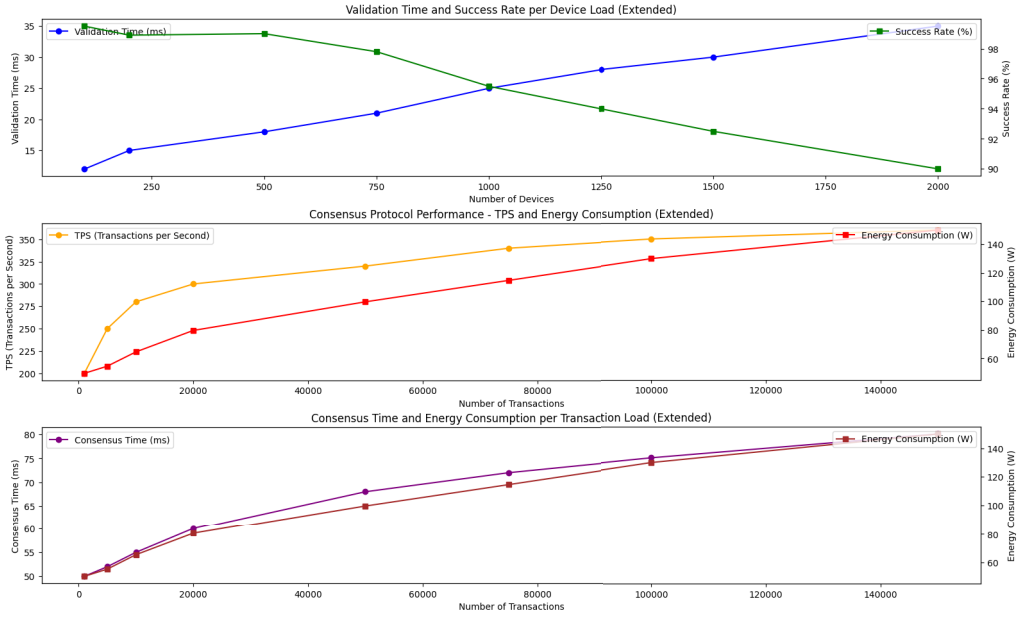


FIGURE 9. System performance in terms of validation time, tps, and power consumption under different device and transaction loads.

TABLE 10. Intrusion detection precision compared between hybrid system and traditional approaches.

A	B	C	D	E	F
Hybrid (AI and Blockchain)	Phishing	95.2	93.5	3.1	1.7
Traditional	Phishing	85.3	80.2	7.8	6.9
Hybrid (AI and Blockchain)	DoS Attack	97.5	96.0	2.5	1.0
Traditional	DoS Attack	88.9	84.5	6.5	4.6
Hybrid (AI and Blockchain)	Unauthorized Access	96.8	95.4	2.3	1.5
Traditional	Unauthorized Access	87.2	83.7	5.9	6.2

Note:

- A: System
- B: Type of Attack
- C: Precision (%)
- D: F1-Score (%)
- E: False Positives (%)
- F: False Negatives (%)

TABLE 11. Response latency compared between hybrid system and traditional approaches.

System	Transaction Type	Response Latency (ms)	Variability (%)
Hybrid (AI and Blockchain)	Device Authentication	15	5
Traditional	Device Authentication	45	10
Hybrid (AI and Blockchain)	Data Transfer	18	4
Traditional	Data Transfer	50	12
Hybrid (AI and Blockchain)	Transaction Validation	20	6
Traditional	Transaction Validation	60	15

when detecting intrusions in various attacks. In the case of spoofing, the hybrid system achieves a precision of 95.2%, while the traditional approach remains at 85.3%. Furthermore, the hybrid system?s F1-Score of 93.5% reflects a superior balance between precision and recall compared to the conventional system?s 80.2%.

The difference is even more notable in DoS attacks, with the hybrid system achieving 97.5% precision versus 88.9% for the traditional approach. This difference is due to the hybrid system?s ability to adapt and detect anomalous patterns using AI algorithms dynamically. In contrast, conventional systems rely on predefined rules that limit their effectiveness against new attack variants.

The hybrid system also excels in unauthorized access attacks, with an precision of 96.8%, surpassing the 87.2% of conventional systems. In addition, the results show a lower rate of false positives and false negatives in the hybrid system, which improves its ability to distinguish between legitimate and malicious traffic, reducing the likelihood of unnecessary network interruptions.

Table 11 compares response latency across different types of transactions. The results indicate that the hybrid system significantly reduces latency compared to traditional systems. In the case of device authentication, the hybrid system has a latency of 15 ms, while the conventional system requires 45 ms. This difference is critical in IoT networks, where response speed is essential to ensure fluid operations.

The hybrid system also demonstrates an improvement in data transfers, with a latency of 18 ms, compared to 50 ms for conventional systems. This improvement is mainly due to edge nodes that allow transaction processing and validation

TABLE 12. Energy consumption and scalability compared between hybrid system and traditional approaches.

System	Number of Connected IoT Devices	Energy Consumption (W)	Scalability (% Efficiency)
Hybrid (AI and Blockchain)	500	100	98%
Traditional	500	150	80%
Hybrid (AI and Blockchain)	1000	150	95%
Traditional	1000	220	75%
Hybrid (AI and Blockchain)	2000	200	92%
Traditional	2000	300	70%

locally, reducing the need to resort to central servers and eliminating associated delays.

Also, in transaction validation, the hybrid system achieves a latency of 20 ms, compared to 60 ms for the traditional approach. This is particularly relevant in applications that require secure and fast transaction validation, such as industrial IoT networks or critical systems where high latency could compromise data security or integrity.

Table 12 compares power consumption and scalability between hybrid and traditional systems under different load scenarios. For 500 connected IoT devices, the hybrid system consumes 100 W, significantly less than the 150 W of conventional approaches. This energy efficiency is maintained even when the number of devices increases to 1000 and 2000, where the hybrid system consumes 150 W and 200 W, respectively, while the traditional approach reaches 300 W in the highest load scenario.

The hybrid system shows significantly higher scalability, maintaining 95% efficiency with 1,000 devices, while traditional systems only reach 75% in the same scenario. This scalability is derived from the combination of blockchain and AI, which allows processing and authentication tasks to be distributed across edge nodes, avoiding overloading central resources and maximizing performance without significantly impacting energy consumption.

As the network grows to 2,000 devices, the hybrid system continues to operate efficiently, with 92% scalability, compared to 70% for conventional systems. This shows that the hybrid system is more energy efficient and can better adapt to large-scale IoT networks without compromising performance.

V. DISCUSSION

The works of Hayder and Abdulkadhim [46] and Wei et al. [47] have pointed out the advances in the use of neural networks and cloud processing. Still, their systems have presented problems related to high latency and high energy consumption. Our work demonstrates that integrating blockchain with AI improves the precision of intrusion detection, reduces latency, and optimizes energy consumption, which addresses the shortcomings observed in these previous studies. The significantly lower false positives and negatives in our approach and the ability of the system to maintain its efficiency under high-load scenarios reveal a crucial innovation in managing security in IoT networks.

Our process included the implementation of smart contracts for device authentication and real-time transaction validation. Unlike traditional approaches that rely on predefined or centralized rules, smart contracts on edge nodes allow the network to distribute processing tasks, minimizing latency and ensuring security from the data origin point. In this sense, the lightweight consensus protocol we designed was crucial to maintain the balance between security and energy efficiency, allowing us to process large volumes of transactions with a controlled energy cost, as observed in the energy consumption results in previous tables [48]. This protocol, combined with deep learning models, facilitated the early detection of abnormal patterns and the prevention of threats in real time. This is essential in IoT networks where threats are constantly evolving.

An essential contribution of this work is the incorporation of specific simulations of internal attacks in IoT environments. These simulations replicate critical scenarios, such as compromised devices generating malicious traffic or attempting to impersonate legitimate devices from within the network. These situations are particularly challenging in IoT, where device heterogeneity and lack of intrinsic trust between them increase vulnerabilities. The simulations complement the inherent limitations of the UNSW-NB15 dataset and provide a robust experimental validation of the hybrid AI- and blockchain-based scheme. This approach enables real-time detection and mitigation of internal anomalous behaviors, significantly improving security and trust in dynamic IoT environments.

Blockchain for device authentication and data integrity provides several advantages over traditional methods, such as SSL/TLS. While SSL/TLS ensures encrypted communication between devices, it relies on centralized CAs, which can become a single point of failure or attack target. Blockchain, in contrast, provides a decentralized mechanism that eliminates this dependency and enhances resilience. Moreover, Blockchain enables the traceability and immutability of transactions, allowing the network to maintain a historical record of all interactions. This ensures accountability and provides an added layer of security against attacks such as spoofing or unauthorized access. While Blockchain complements secure communication protocols like SSL/TLS, it addresses the broader trust model challenges inherent in heterogeneous IoT networks, where devices may not inherently trust each other.

The proposed hybrid AI and blockchain model demonstrates high adaptability to evolving IoT environments. This adaptability is crucial in sudden increases in device connections or traffic surges, which are common in real-world applications such as smart cities or industrial IoT networks. The system's edge-cloud architecture allows for dynamic workload distribution. Edge nodes handle local data processing during traffic surges, prioritizing latency-sensitive tasks while less critical operations are deferred to the cloud.

This mechanism effectively mitigates the risk of network congestion and ensures a timely response to threats.

PoA consensus protocol further enhances scalability by limiting the number of validators to trusted edge nodes. This minimizes the computational overhead associated with transaction validation, enabling the system to accommodate a higher volume of transactions and additional device connections without significant degradation in performance.

Additionally, the AI models employed in the system are equipped with online learning capabilities, allowing them to adapt to emerging traffic patterns and anomalies in real-time. This ensures robust intrusion detection even under conditions of high traffic variability. Autoencoders, in particular, dynamically adjust their anomaly detection thresholds based on updated training data, maintaining detection precision during periods of network volatility.

The results show that the system achieves a precision of up to 97.5% in detecting DoS attacks and a reduction of latency in device authentication to 15 ms, outperforming the metrics obtained in traditional systems. These results are consistent with the initial hypothesis that a hybrid AI and blockchain approach would improve security and efficiency in IoT networks. Furthermore, the system power consumption under moderate load (with 1000 devices) remained at 150 W, representing a considerable improvement over the 220 W of traditional systems, highlighting the importance of processing distribution among edge nodes.

However, one of the essential aspects of the discussion is the limitation of energy consumption in large-scale IoT networks. Although our system proved to be more efficient than traditional solutions, it is essential to note that energy consumption tends to increase proportionally with the number of devices and transactions. This suggests that, while the solution is viable for medium- or moderate-sized networks, the energy impact on extensive networks might require additional optimizations, such as using advanced energy optimization techniques or introducing emerging technologies such as quantum computing or low-energy blockchain [19]. This is one of the limitations that should be addressed in future work.

The implications of energy efficiency become even more critical in large-scale IoT implementations, such as smart cities or industrial automation. High resource consumption by security models can limit the practical deployment of such systems. While our lightweight consensus protocol addresses this significantly, the trade-off lies in balancing system security with the scalability of energy-efficient operations. Deploying the system at scale would require careful consideration of infrastructure capacity, including integrating energy-saving technologies such as dynamic voltage scaling or hardware accelerators. These strategies could optimize performance while minimizing the energy footprint, making the system viable for widespread adoption.

Another key limitation concerns the system's scalability. Although the proposed solution shows 95% scalability with 1000 connected devices, efficiency starts to decrease when facing extreme loads, suggesting that performance challenges might arise in extensive IoT applications, such as large-scale smart cities or critical infrastructures with millions of devices. While our system is more efficient than traditional approaches, the workload associated with validating transactions over the blockchain can become a bottleneck, especially when processing multiple transactions simultaneously on low-capacity devices.

Furthermore, our study assumes IoT devices have sufficient processing power to run real-time validation and data analysis algorithms. This may not always be applicable to IoT devices with minimal resources, such as simple sensors that cannot support complex computational processes [49]. In scenarios with straightforward devices, it would be necessary to implement a lighter processing mechanism or delegate all processing to edge nodes or the cloud, which could sometimes compromise latency.

VI. CONCLUSION

The proposed hybrid solution demonstrates significant advancements in intrusion detection precision, scalability, and energy efficiency for IoT networks. Combining blockchain for secure authentication with adaptive AI models for anomaly detection, the system achieves a robust and scalable architecture suited for diverse applications such as smart cities, Industry 4.0, and precision agriculture.

The solution effectively addresses key limitations of traditional approaches, such as high latency and inefficiency in energy use. It achieves an authentication latency of 15 ms and energy consumption of 150 W in networks with up to 1000 devices, highlighting its viability for resource-constrained IoT environments. Integrating lightweight consensus protocols and decentralized processing ensures security and scalability without compromising performance.

While the results validate the system's effectiveness in medium-sized networks and controlled environments, certain limitations remain. These include potential challenges in highly dynamic scenarios and the need to further optimize blockchain protocols for large-scale networks. Future research should explore emerging technologies, such as low-energy blockchain techniques and federated learning, to enhance energy efficiency and adapt to more demanding environments. The findings underscore the importance of integrating blockchain and AI in IoT security frameworks, paving the way for robust, scalable, and energy-efficient solutions in modern IoT ecosystems.

One aspect that deserves future exploration is the comparison of the hybrid blockchain- and AI-based solution with advanced approaches such as federated learning. This paradigm, prioritizing distributed training and data privacy, could offer a complementary perspective for IoT environments where data confidentiality is critical. Additionally, future research should include a comprehensive evaluation of the system's performance against models based on Generative Adversarial Networks (GANs), which have

proven effective in generating synthetic attack patterns and enhancing the robustness of intrusion detection systems. This comparison will provide deeper insights into the hybrid approach's advantages and limitations in scenarios involving dynamic and evolving threats. These advanced methodologies will extend the current study's scope and contribute to a more comprehensive understanding of security mechanisms in heterogeneous IoT networks.

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